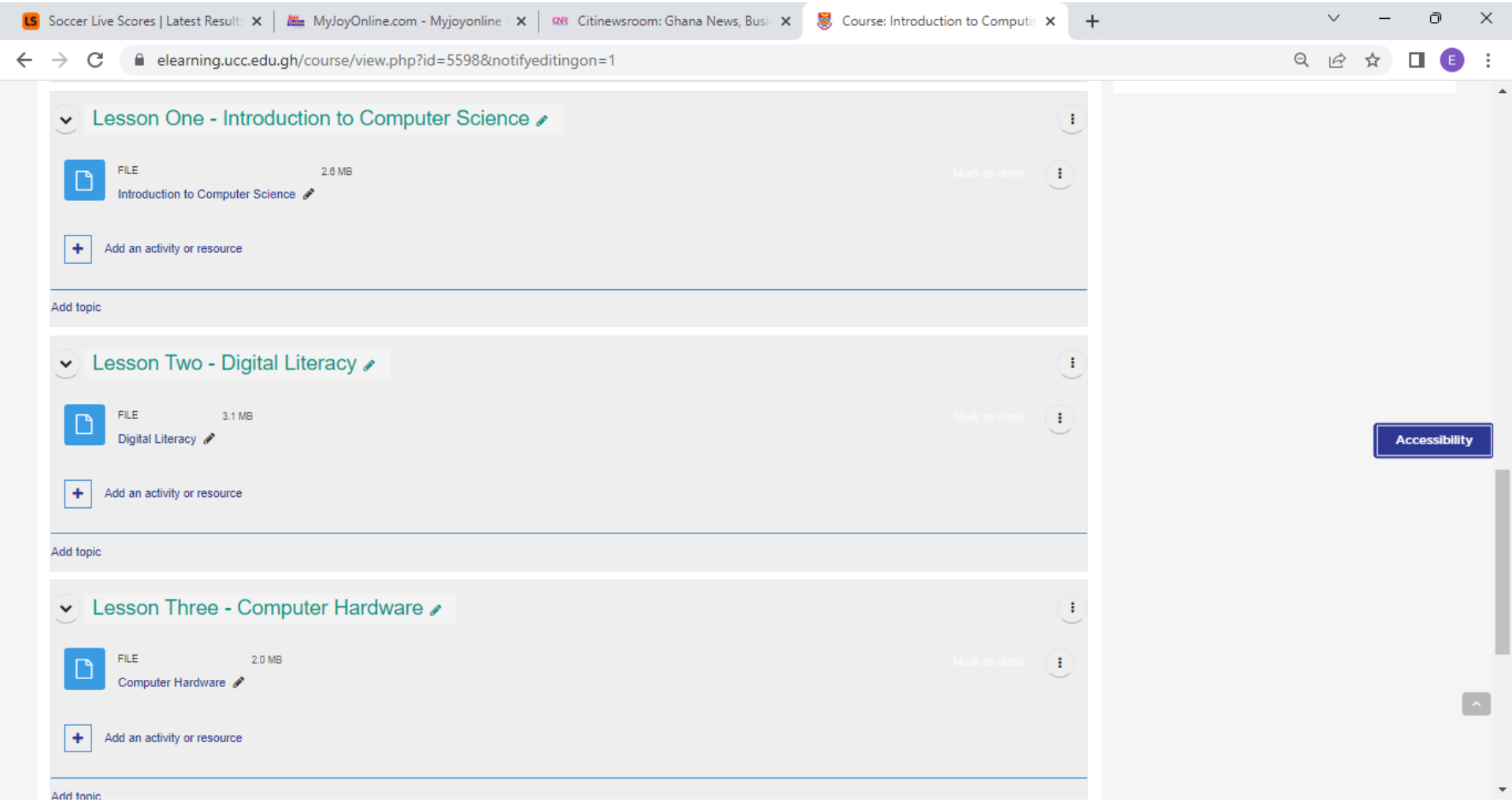
The Internet & World Wide Web

Elliot Attipoe


Soccer Live Scores | Latest Result
MyJoyOnline.com - Myjoyonline
Citinewsroom: Ghana News, Busi
Course: Introduction to Computi
elearning.ucc.edu.gh/course/view.php?id=5598&notifyeditingon=1
Lesson One - Introduction to Computer Science
FILE 2.6 MB
Introduction to Computer Science
Add an activity or resource
Add topic
Lesson Two - Digital Literacy
FILE 3.1 MB
Digital Literacy
Add an activity or resource
Add topic
Lesson Three - Computer Hardware
FILE 2.0 MB
Computer Hardware
Add an activity or resource
Add topic
Accessibility

Internet Usage Stats

WORLD INTERNET USAGE AND POPULATION STATISTICS 2023 Year Estimates						
World Regions	Population (2022 Est.)	Population % of World	Internet Users 31 Dec 2021	Penetration Rate (% Pop.)	Growth 2000-2023	Internet World %
Africa	1,394,588,547	17.6 %	601,940,784	43.2 %	13,233 %	11.2 %
Asia	4,352,169,960	54.9 %	2,916,890,209	67.0 %	2,452 %	54.2 %
Europe	837,472,045	10.6 %	747,214,734	89.2 %	611 %	13.9 %
Latin America / Carib.	664,099,841	8.4 %	534,526,057	80.5 %	2,858 %	9.9 %
North America	372,555,585	4.7 %	347,916,694	93.4 %	222 %	6.5 %
Middle East	268,302,801	3.4 %	206,760,743	77.1 %	6,194 %	3.8 %
Oceania / Australia	43,602,955	0.5 %	30,549,185	70.1 %	301 %	0.6 %
WORLD TOTAL	7,932,791,734	100.0 %	5,385,798,406	67.9 %	1,392 %	100.0 %

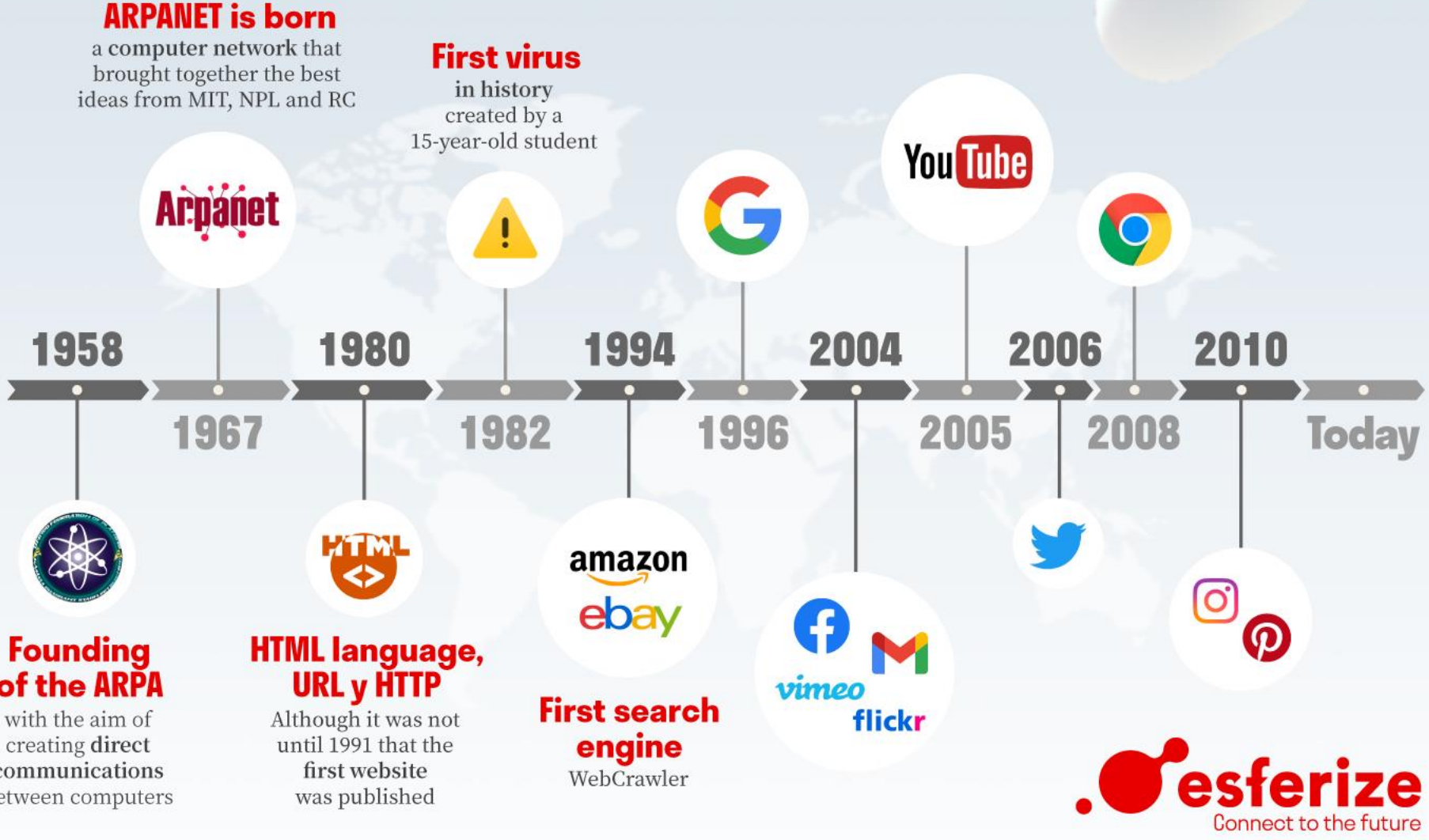
Source: https://www.internetworldstats.com/stats.htm

Internet Users Statistics for Africa

(Africa Internet Usage, 2022 Population Stats and Facebook Subscribers)

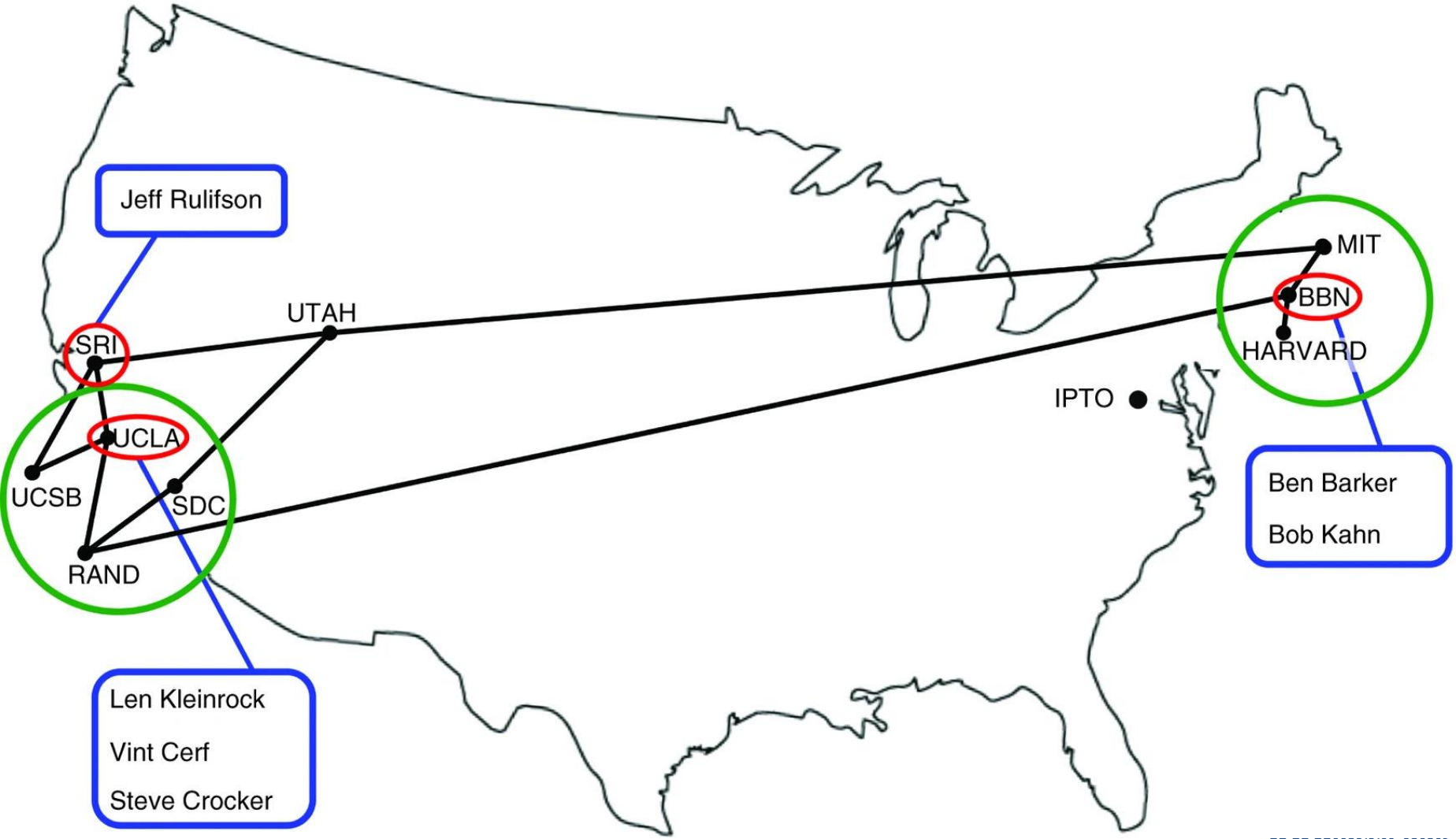
AFRICA 2022 POPULATION AND INTERNET USERS STATISTICS						
AFRICA	Population (2022 Est.)	Internet Users 31-DEC-2000	Internet Users 31-DEC-21	Internet Penetration	Internet Growth % 2000 - 2021	Facebook subscribers 30-APRIL-22
Algeria	45,150,879	50,000	37,836,425	83.8 %	50,756 %	26,291,400
Angola	34,592,611	30,000	8,980,670	26.0 %	29,835 %	2,875,600
Benin	12,653,644	15,000	3,801,758	30.0 %	25,245 %	1,686,800
Botswana	2,429,926	15,000	1,247,000	51.3 %	7,493 %	1,191,300
Ghana	32,154,245	30,000	14,767,818	45.9 %	49,126 %	9,163,200
Guinea	13,734,762	8,000	2,551,672	18.6 %	31,795 %	2,446,700
Guinea-Bissau	2,046,008	1,500	900,000	44.0 %	16,566 %	900,000
Kenya	55,752,020	200,000	46,870,422	85.2 %	23,335 %	12,445,700
Nigeria	211,400,708	200,000	154,301,195	73.0 %	101,484 %	31,860,000
Reunion (FR)	901,686	130,000	608,000	67.4 %	367 %	608,000
Rwanda	13,276,513	5,000	5,981,638	45.1 %	119,532 %	806,200
Sao Tome & Principe	223,368	6,500	63,864	28.6 %	882 %	60,800
Senegal	17,196,301	40,000	9,749,527	56.7 %	24,273 %	3,802,000

Evolution of the Internet



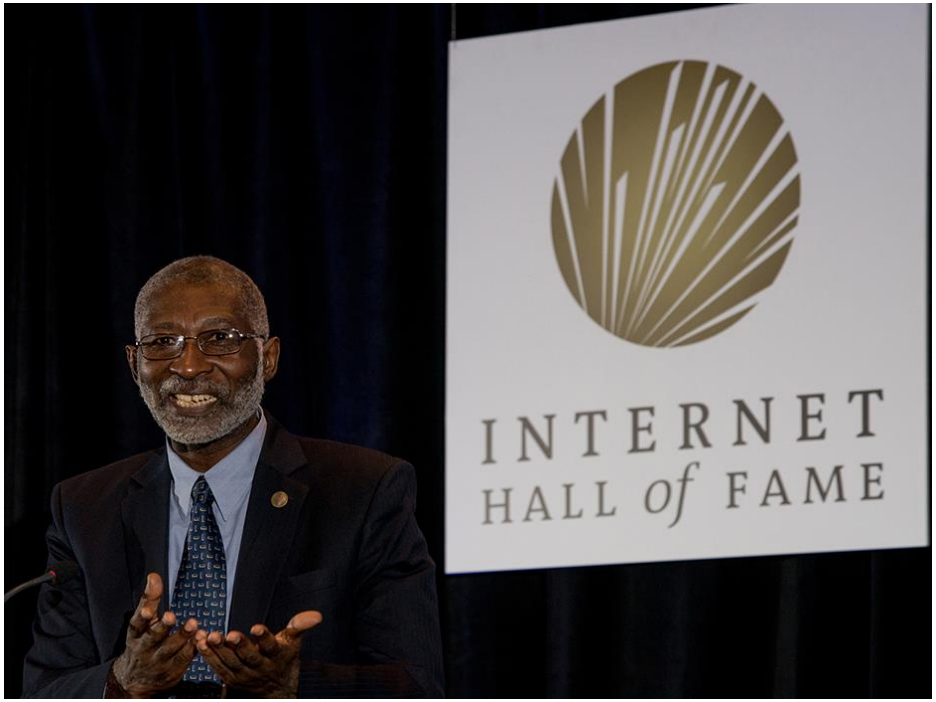
esferize
Connect to the future

ARPANET


Jeff Rulifson
SRI
UTAH
UCLA
UCSB
SDC
RAND
Len Kleinrock
Vint Cerf
Steve Crocker
IPTO
MIT
BBN
HARVARD
Ben Barker
Bob Kahn

History of Internet in Ghana

- Ghana is known to be the first country in West Africa to be connected to the Internet in **1994.**
- Prof. Nii Narku Quaynor is attributed to have brought internet to Ghana.



Networking Fundamentals

- Computer network:
  - Two or more computers connected together

- Benefits of a network:
  - Sharing resources
  - Transferring files

Types of Network

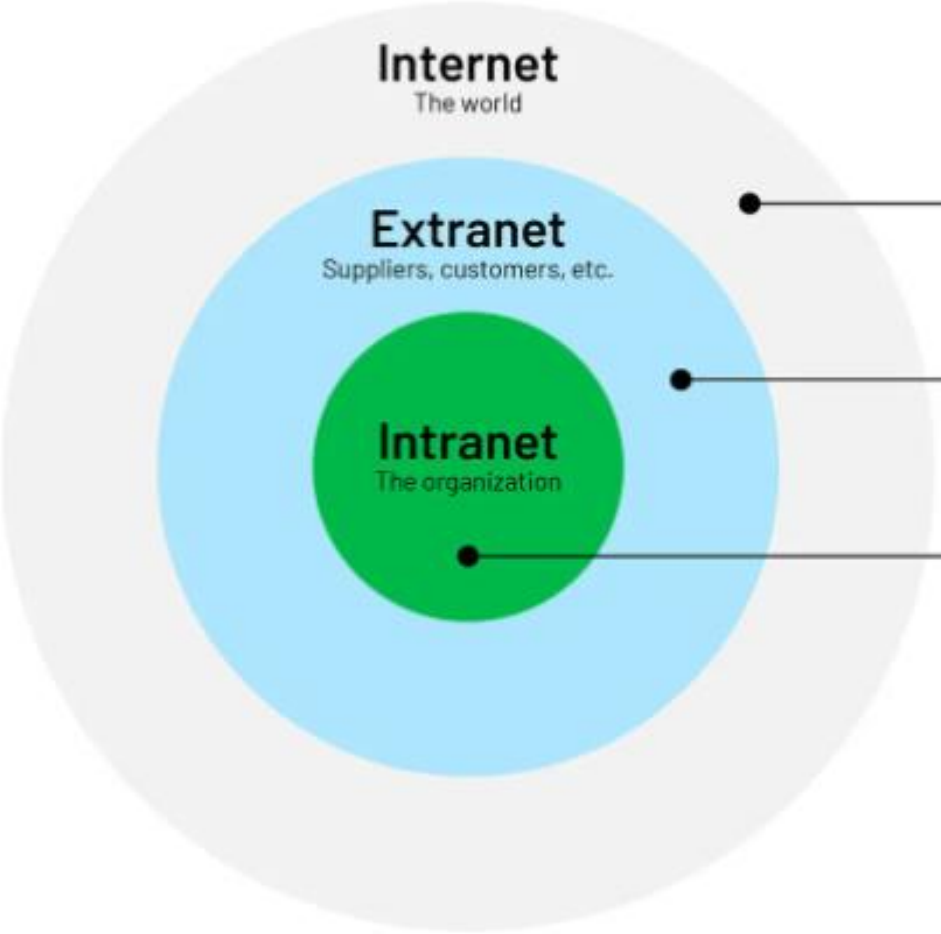
- A **local area network** (**LAN**) is a network that connects computers and devices in a limited geographical area such as a home, school, office building.
- A **wide area network** (**WAN**) is a network that covers a large geographic area (such as a city, country, or the world) using a variety of wired and wireless transmission media.

Other types of Network

- **A *Metropolitan Area Network* (*MAN*)** is a high-speed network that connects local area networks in a metropolitan area, such as a city or town, and handles the bulk of communications activity across that region.
- **Campus Area Network (CAN)** is a computer network that spans a limited geographic area.
- A **Personal Area Network** (**PAN**) is a network that connects computers and devices in an individual's workspace using wired and wireless technology. Devices include smartphones, digital cameras, printers, and more.

What is the relationship between WAN and Internet?

Internet vs Intranet vs Extranet



The **internet** creates connections between computers around the world.

An **extranet** creates connections beyond (or outside) an organization.

An **intranet** creates connections inside an organization.

Parameter	Internet	Intranet	Extranet
TYPE	Public	Private	Private
OBJECTIVE	Information sharing across the globe.	Information sharing between staff/employees.	Information sharing between different organizations.
POSSESSION	By none.	By single organization.	By single/multiple organizations.
ACCESSIBILTY	Anyone	Staff/Employees	Limited access by some other organizations.
DATA SECURITY	Not secured	Secured	Secured
NETWORK AREA	Large	Small	Small
AUTHENTICATION	Not required	Required	Required
COMMUNICATION PROTOCOLS	TCP/IP	TCP/IP	TCP/IP

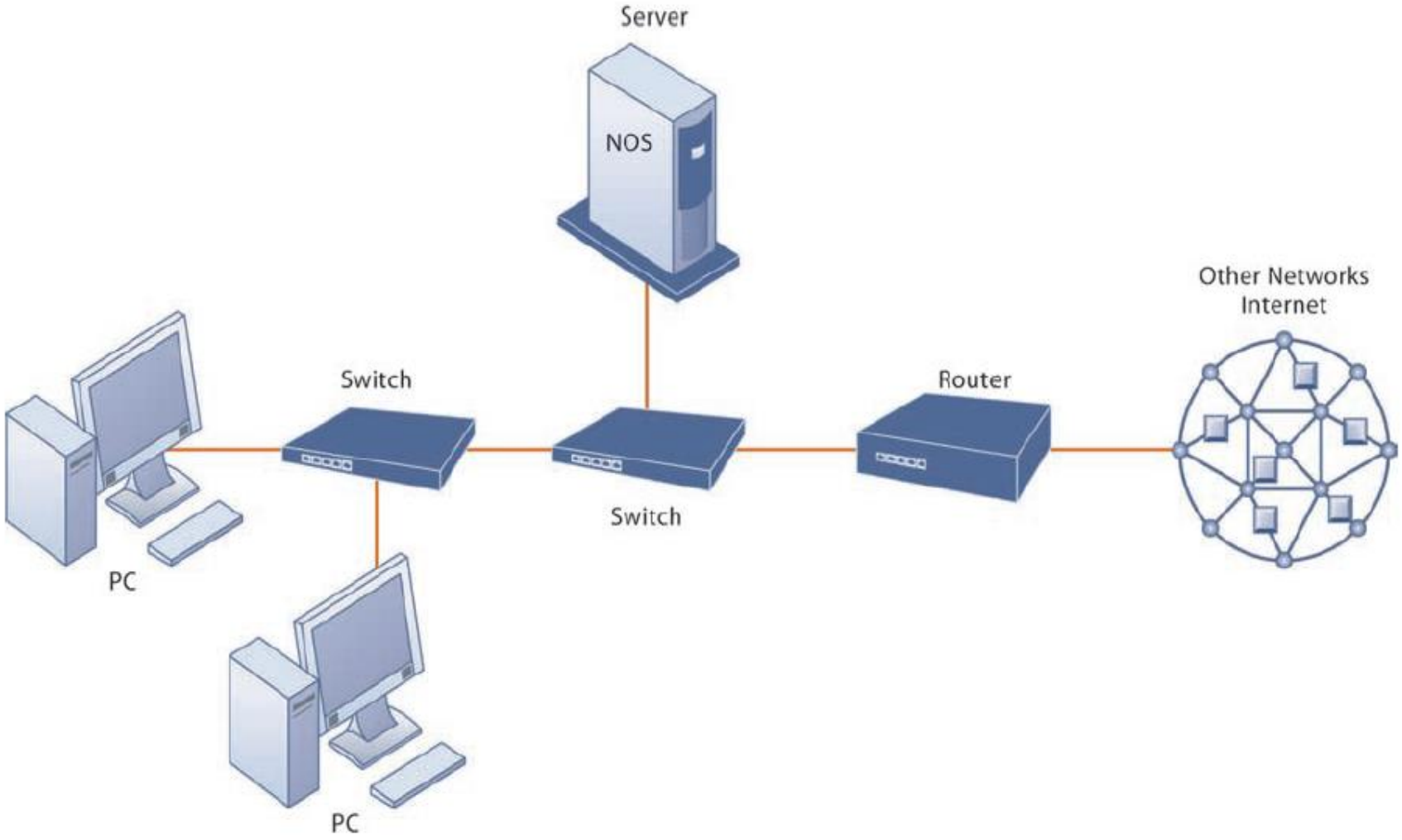
Intranet/Internet Services

- Electronic mails
- Discussion groups/ Newsgroups
- Chatting and Instant messaging
- Internet Telephony (VoIP)
- File Transfer Protocol (FTP)
- Video conferencing
- World wide web (Websites)

Components of computer network

1. Server
2. Transmission media
3. Network Interface Card
4. Switch/Hub
5. Router
6. PC/Client/work station/node

Computer Network



Wireless network



Servers



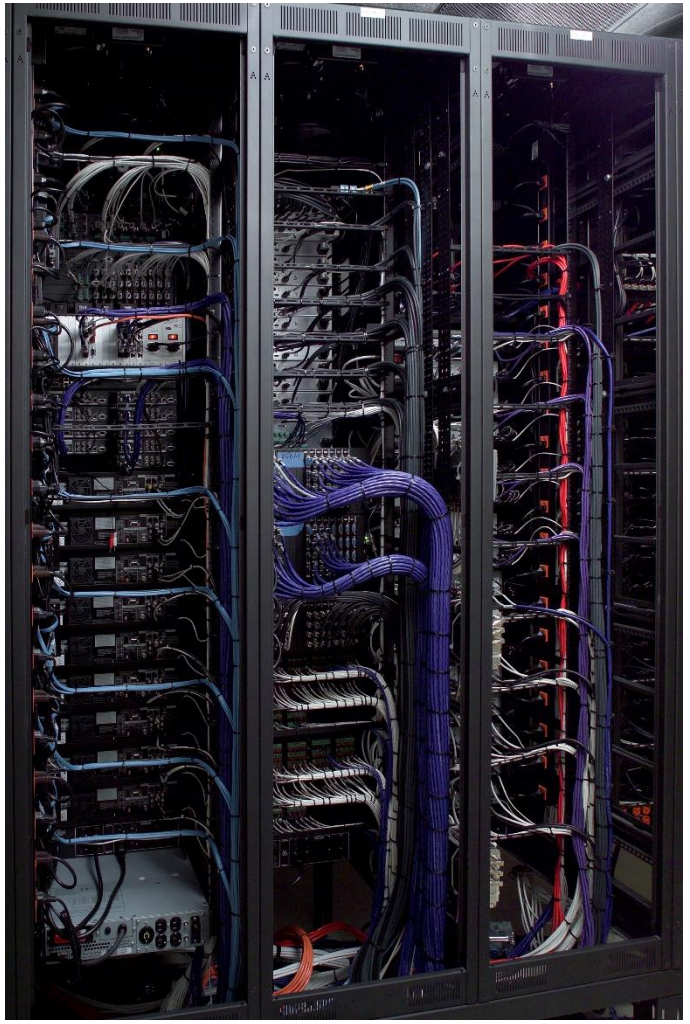


Servers

- Servers are computers that hold shared files, programs, and the network operating system.
- Servers provide access to network resources to all the users of the network.
- There are many different kinds of servers, and one server can provide several functions.
- For example, there are file servers, print servers, mail servers, communication servers, database servers, fax servers and web servers, to name a few.

Rackmount Servers

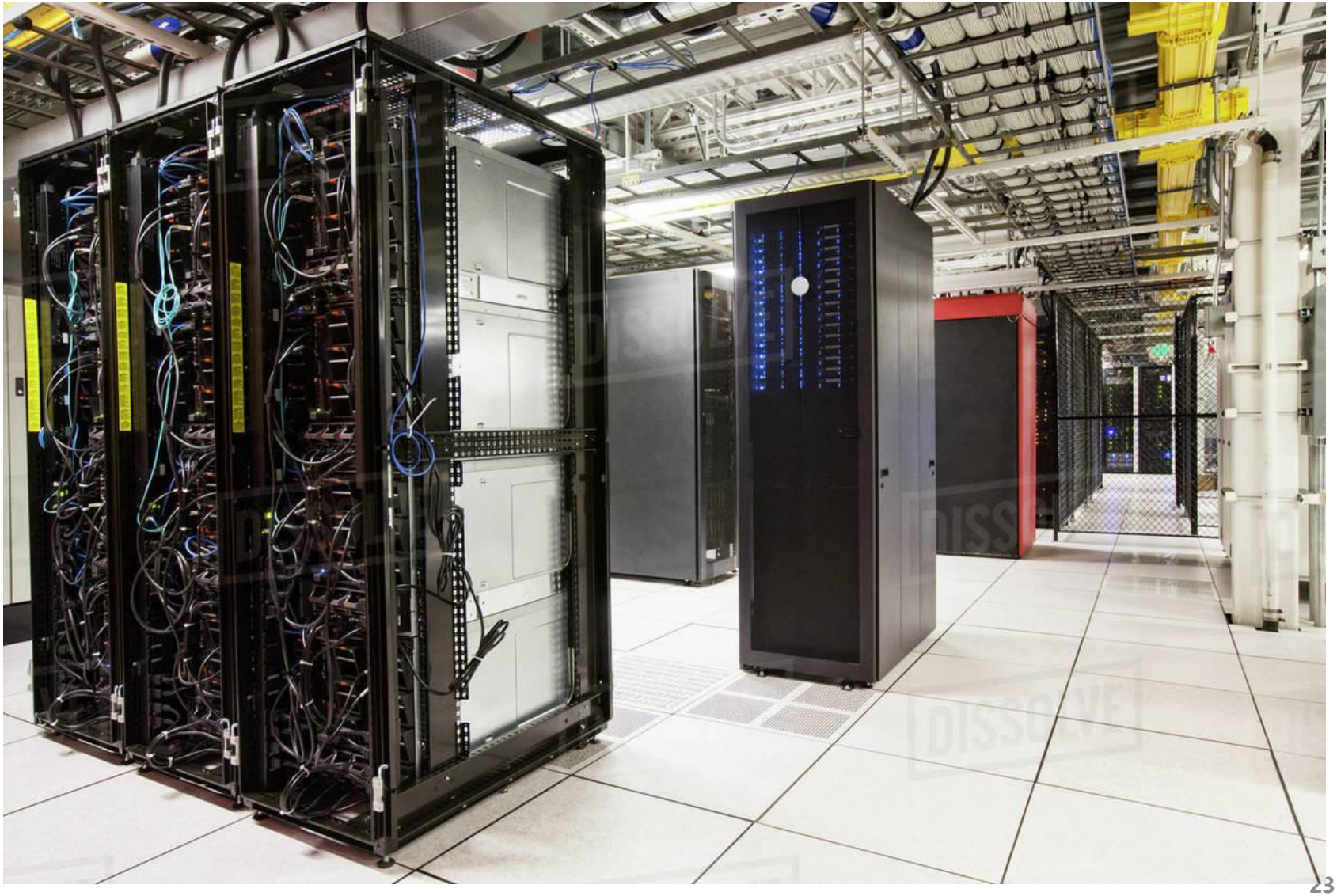




Data Center



Data Center



Google Data Center Locations



Microsoft Azure Data Centers



* Three Azure Government region locations undisclosed

Cloud Computing Platforms



Network Interface Card





RJ45 + Network Interface Card





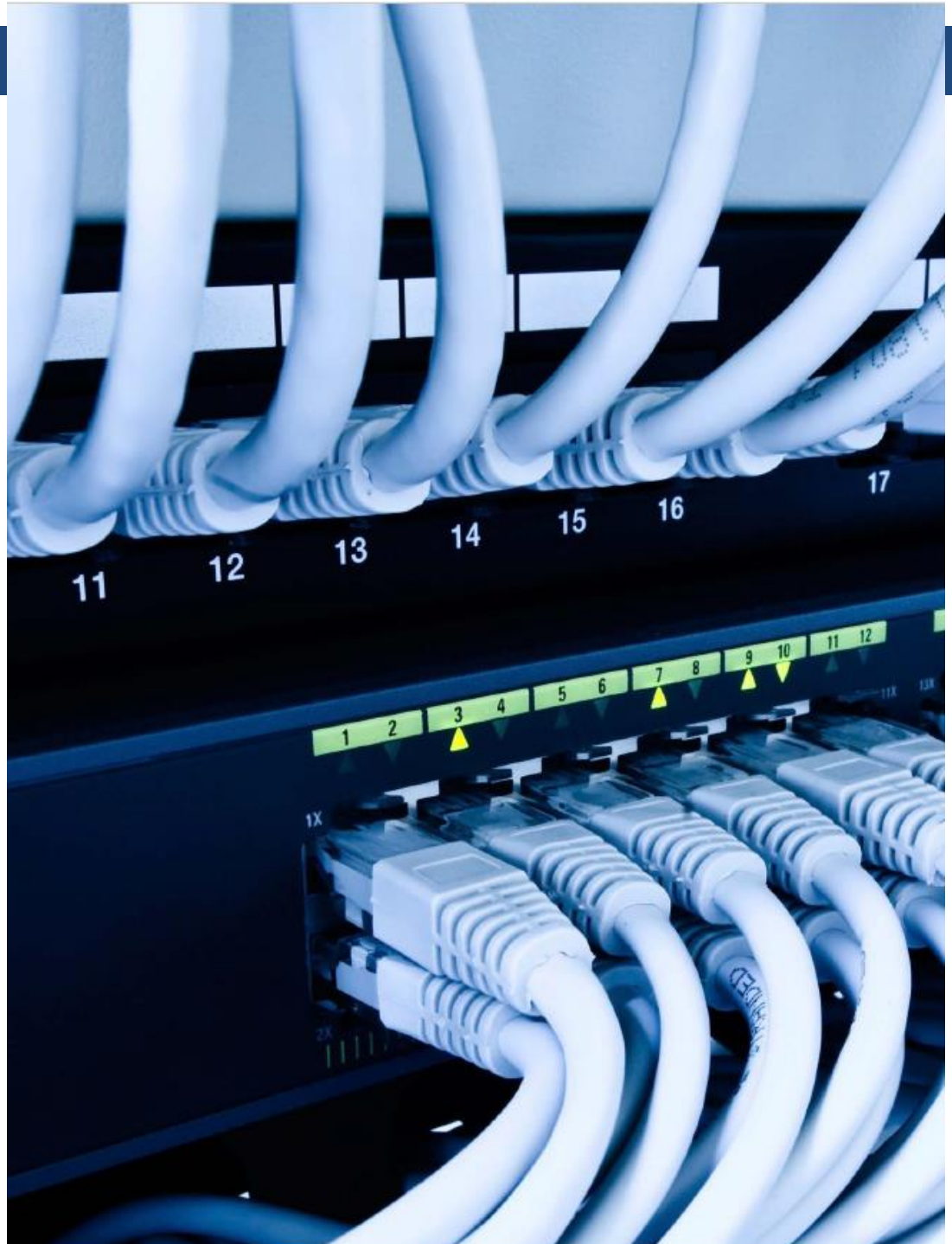
Network Interface Card

- Network Interface Card is a communications device that enables a computer or device that does not have built-in networking capability to access a network.

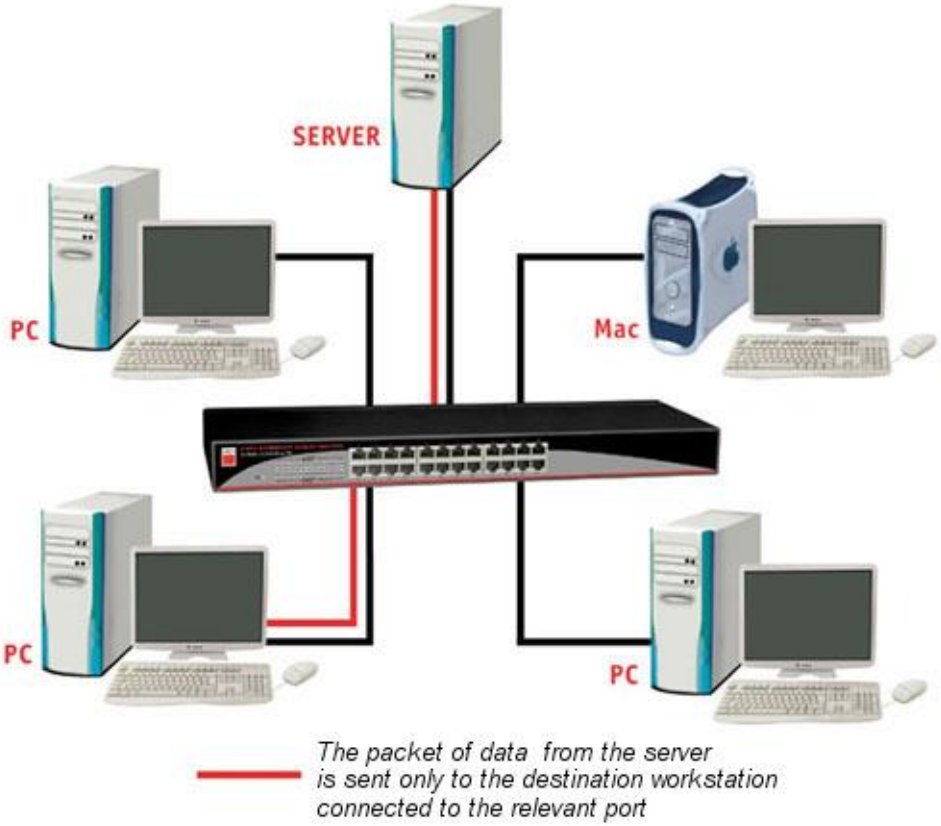
- The network card coordinates the transmission and receipt of data, instructions, and information to and from the computer or device containing the network card.

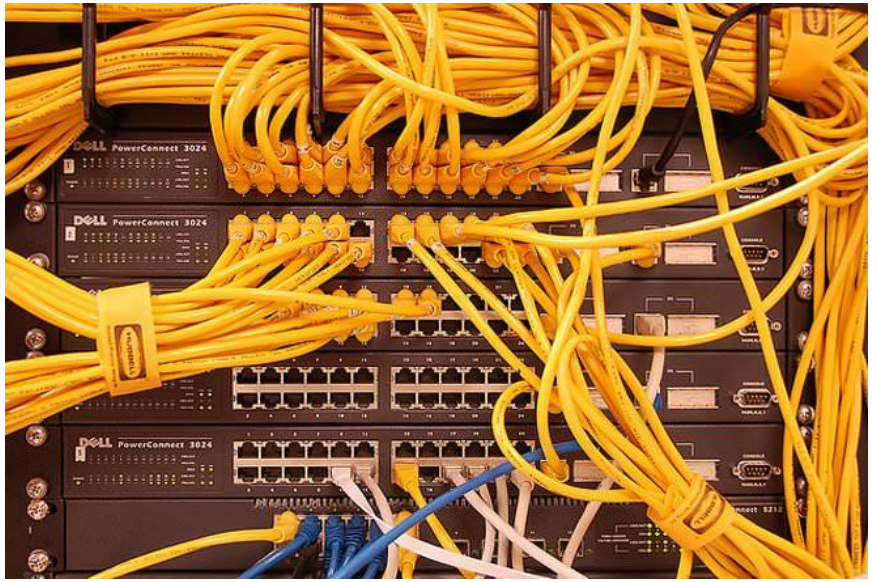
Hub/Switch




11
12
13
14
15
16
17
1 2
3 4
5 6
7 8
9 10
11 12
1X

Computers connected to a switch





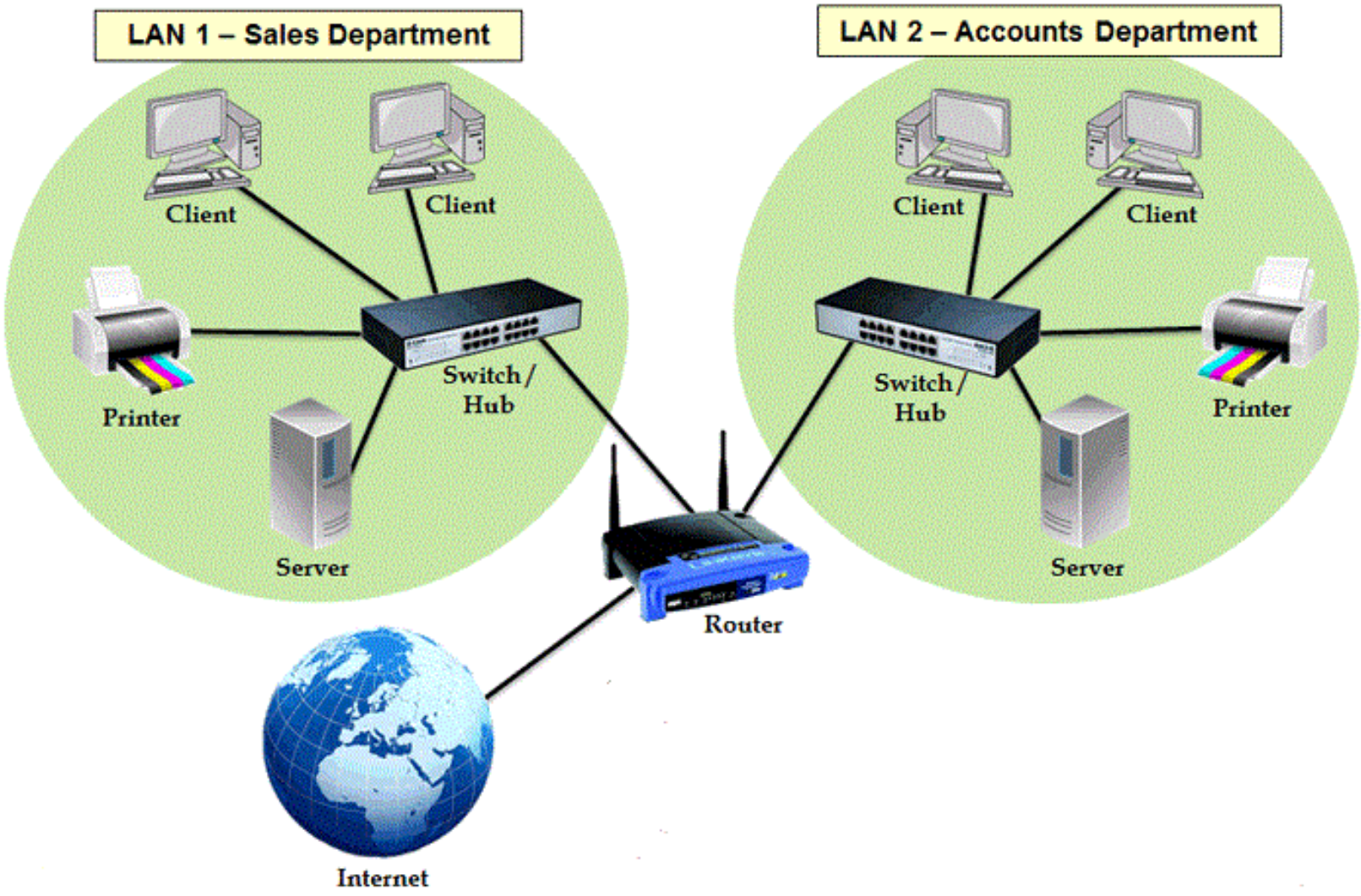
Computer Lab



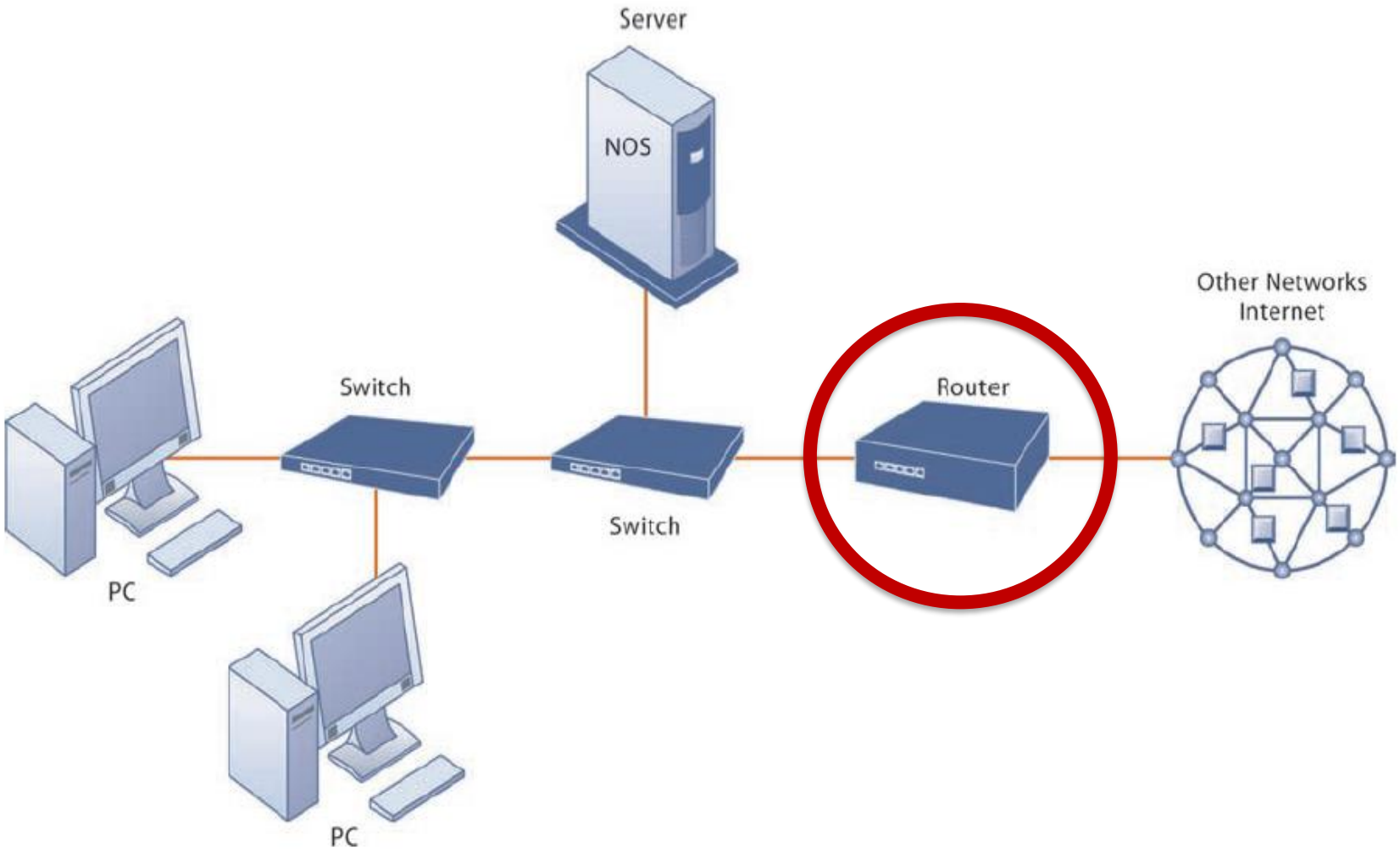
Hub and switches

- A hub or switch is a device that **provides a central point for cables in a network.**
- Larger networks typically use a hub, while smaller networks use a switch.
- Hubs are very simple devices that connect network components, sending a packet of data to all other connected devices.
- A switch has more intelligence than a hub and can filter and forward data to a specified destination on the network.

Router


LAN 1 – Sales Department
LAN 2 – Accounts Department
Client
Client
Client
Client
Printer
Switch / Hub
Switch / Hub
Printer
Server
Server
Router
Internet

Computer network



Router


wireless access point
is center of
wireless network
wired network
Internet
high-speed Internet connection

Router

- A router is a communications device that **connects multiple computers or other routers together** and transmits data to its correct destination on a network.
- A router forwards data packets between computer networks.
- Routers perform the traffic directing functions on the Internet.
- A router can be used on a network of any size.

Transmission Media

- Transmission media consist of materials or substances capable of carrying one or more communications signals.

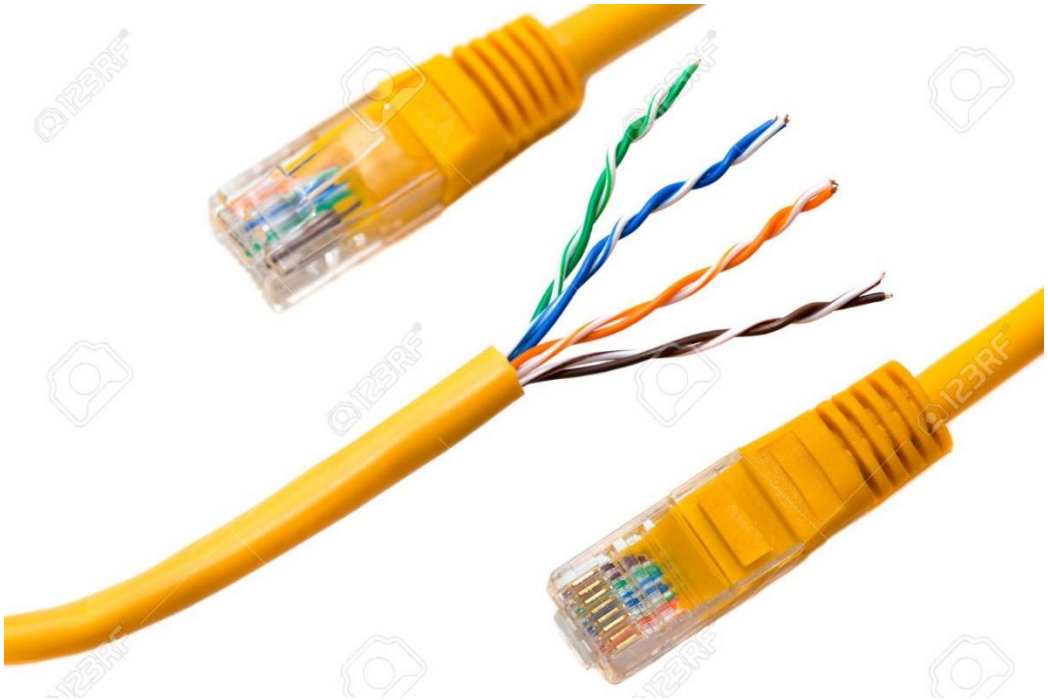
- When you send data from a computer or mobile device, the signal that carries the data
may travel over various transmission media.

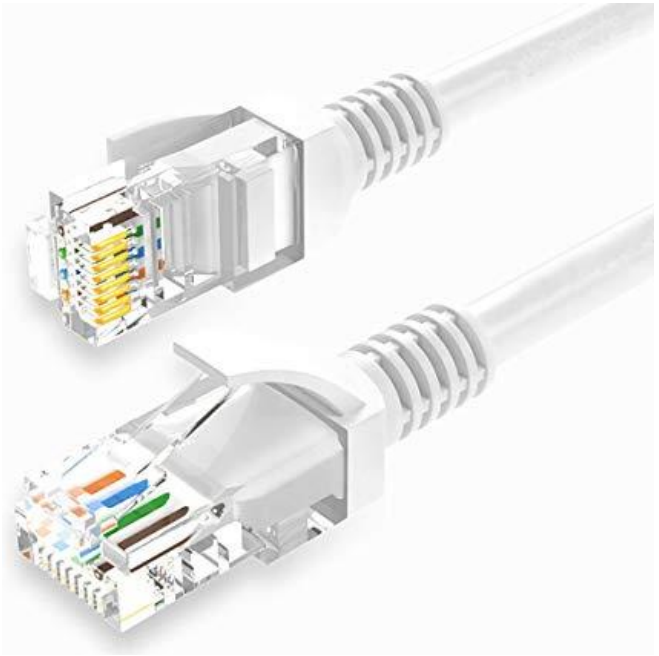
- Classification
  - Physical Transmission Media
  - Wireless Transmission Media

Table 10-3 Transfer Rates for Physical Transmission Media Used in LANs

Type of Cable and LAN	Maximum Transfer Rate
Twisted-Pair Cable	
• 10Base-T (Ethernet)	10 Mbps
• 100Base-T (Fast Ethernet)	100 Mbps
• 1000Base-T (Gigabit Ethernet)	1 Gbps
• Token ring	4 Mbps to 16 Mbps
Coaxial Cable	
• 10Base2 (ThinWire Ethernet)	10 Mbps
• 10Base5 (ThickWire Ethernet)	10 Mbps
Fiber-Optic Cable	
• 10Base-F (Ethernet)	10 Mbps
• 100Base-FX (Fast Ethernet)	100 Mbps
• FDDI (Fiber Distributed Data Interface) token ring	100 Mbps
• Gigabit Ethernet	1 Gbps
• 10-Gigabit Ethernet	10 Gbps
• 40-Gigabit Ethernet	40 Gbps
• 100-Gigabit Ethernet	100 Gbps

Twisted Pair (Ethernet wire cable)

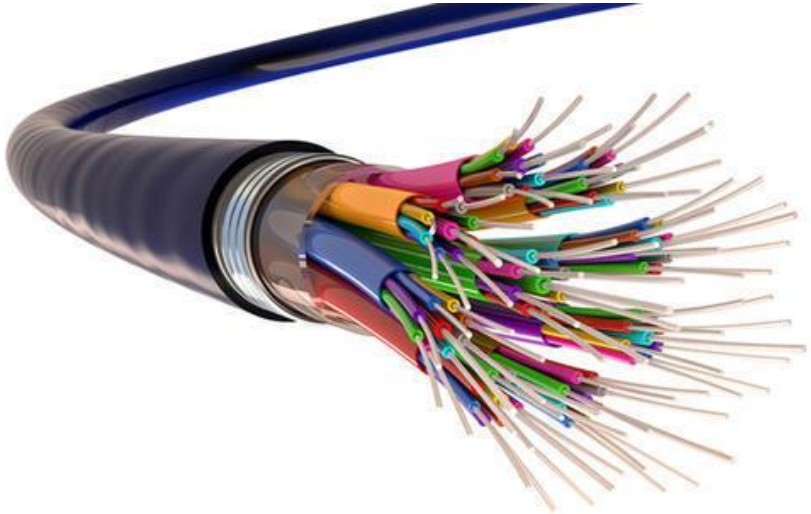


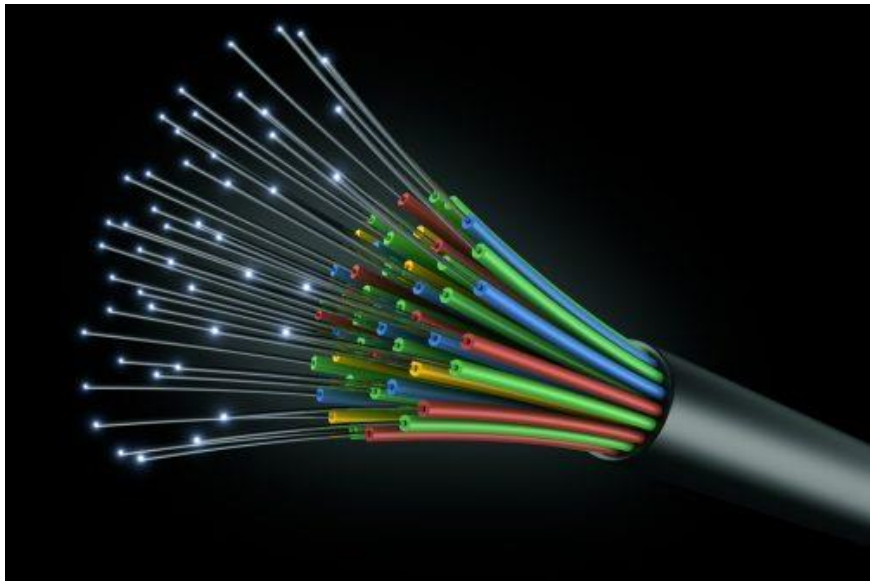


Coaxial Cable

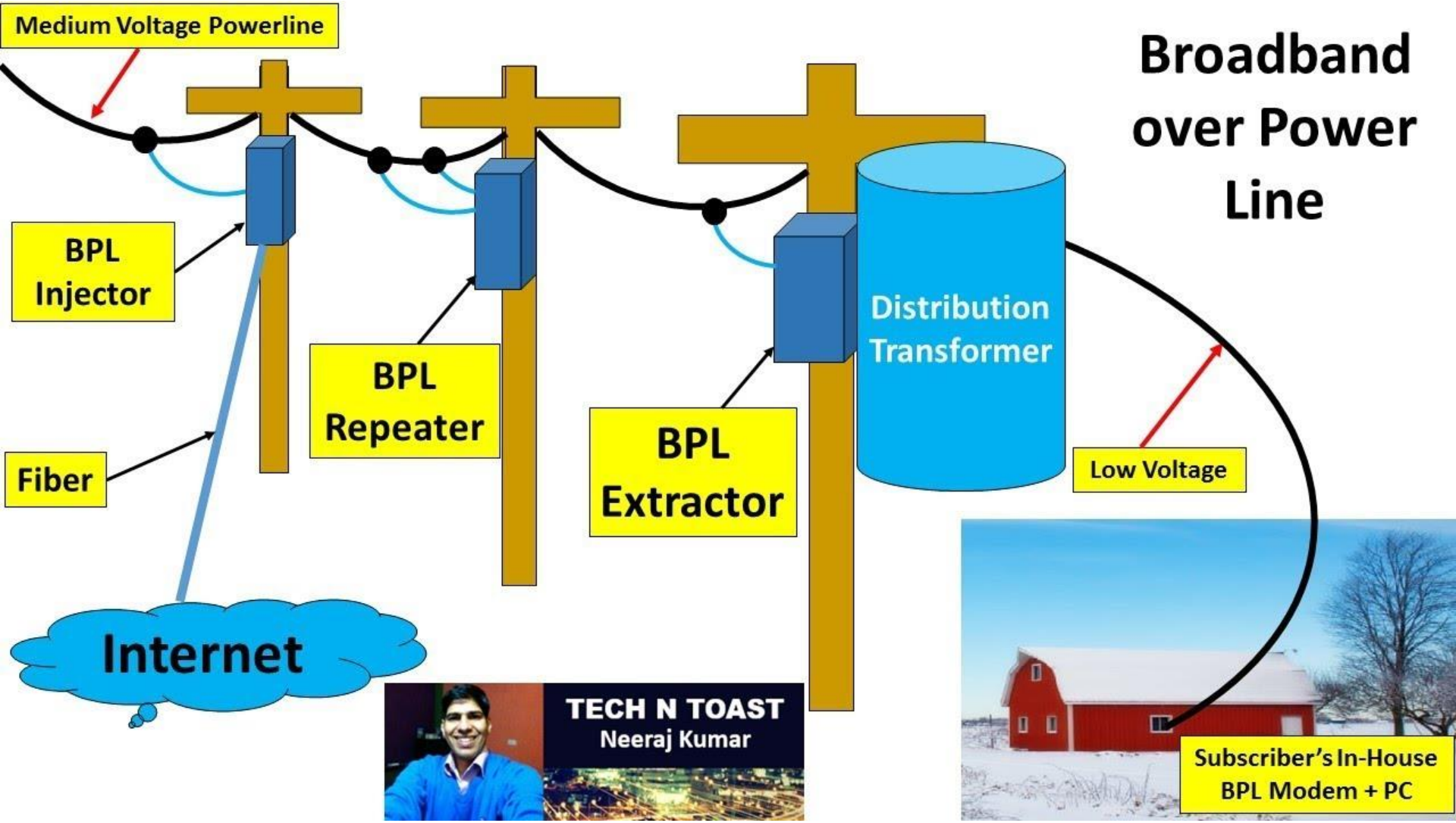


Fiber Optics Cable





Broadband over power line



Wireless Transmission Media

- Wireless transmission media send communications signals through the air or space.

- Many users opt for wireless transmission media because it is more convenient than installing cables.

Table 10-4 Wireless Transmission Media Transfer Rates

Medium		Maximum Transfer Transmission Rate
Infrared		115 Kbps to 4 Mbps
Broadcast radio	• Bluetooth	1 Mbps to 24 Mbps
	• 802.11b	11 Mbps
	• 802.11a	54 Mbps
	• 802.11g	54 Mbps
	• 802.11n	300 Mbps
	• 802.11ac	500 Mbps to 1 Gbps
	• 802.11ad	up to 7 Gbps
	• UWB	110 Mbps to 480 Mbps
Cellular radio	• 2G	9.6 Kbps to 144 Kbps
	• 3G	144 Kbps to 3.84 Mbps
	• 4G	Up to 100 Mbps
Microwave radio		10 Gbps
Communications satellite		2.56 Tbps

Components of computer network

1. Server
2. Transmission media
3. Network Interface Card
4. Switch/Hub
5. Router
6. PC/Client/work station

What is the difference between the Internet and the web?

Internet vs WEB

- The **Internet** is a global network of networks while the **Web** is a collection of information that is accessed via the Internet.
- The **Internet** is infrastructure while the **Web** is served on top of that infrastructure.

Internet vs Web

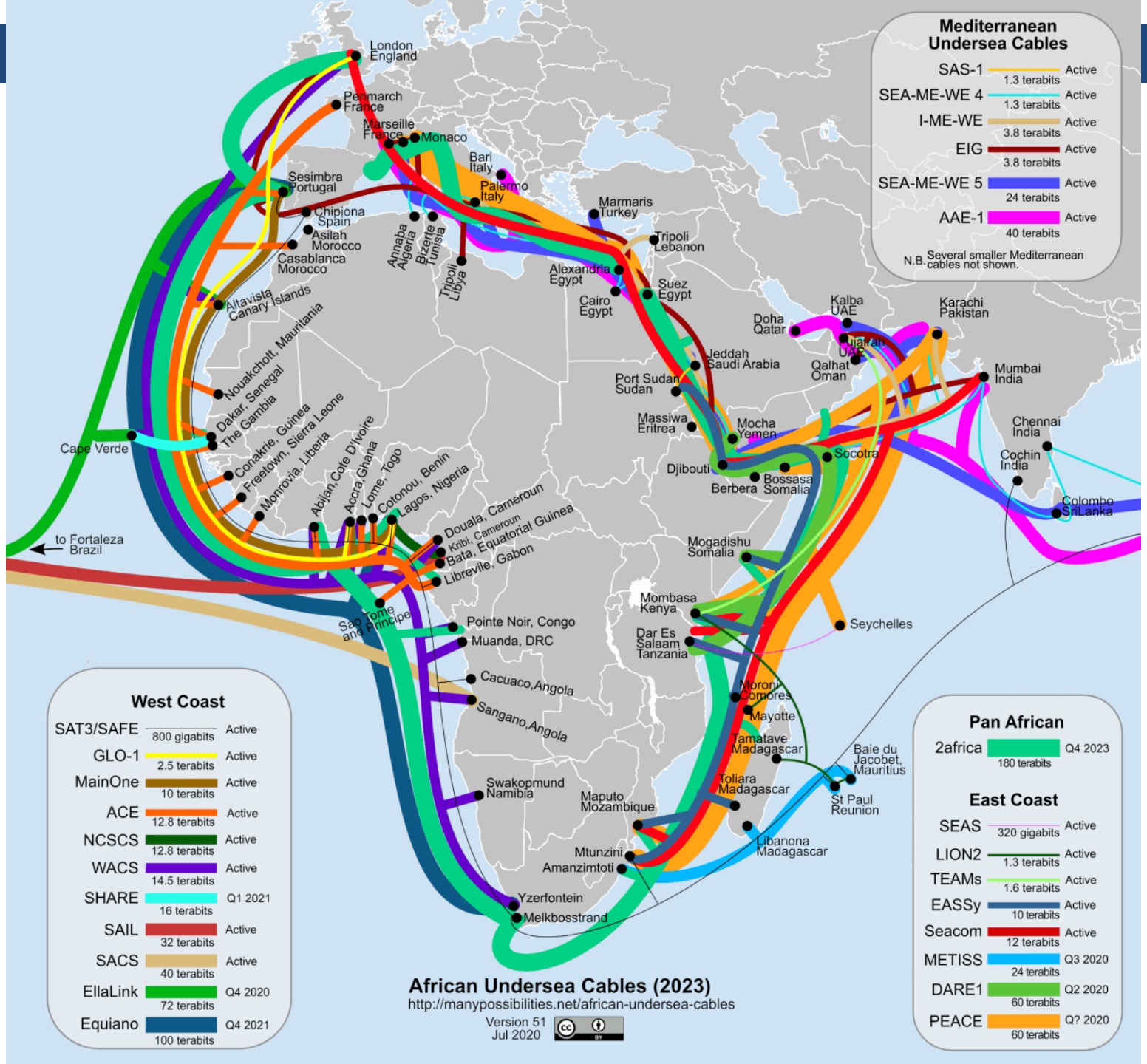
- The Internet contains physical components
  - computers
  - networks
  - services
- The Internet connects thousands of computers across the world, but it is the web that allows communication to occur
- Web - abstraction and common set of services on top of the Internet

Evolution of the Web

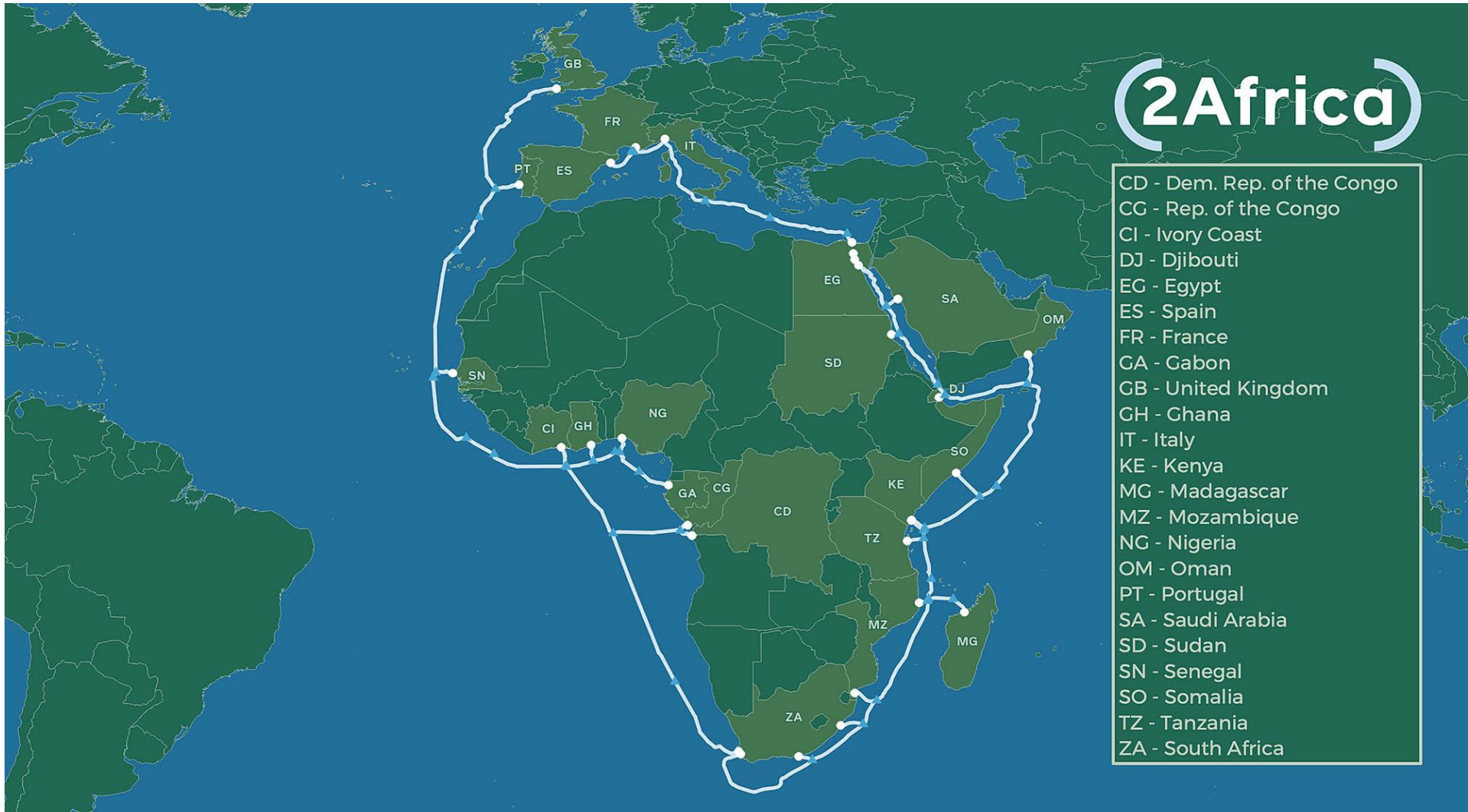
- Sir Tim Berners-Lee is famously credited for the invention of the World Wide World in the early **1989**.



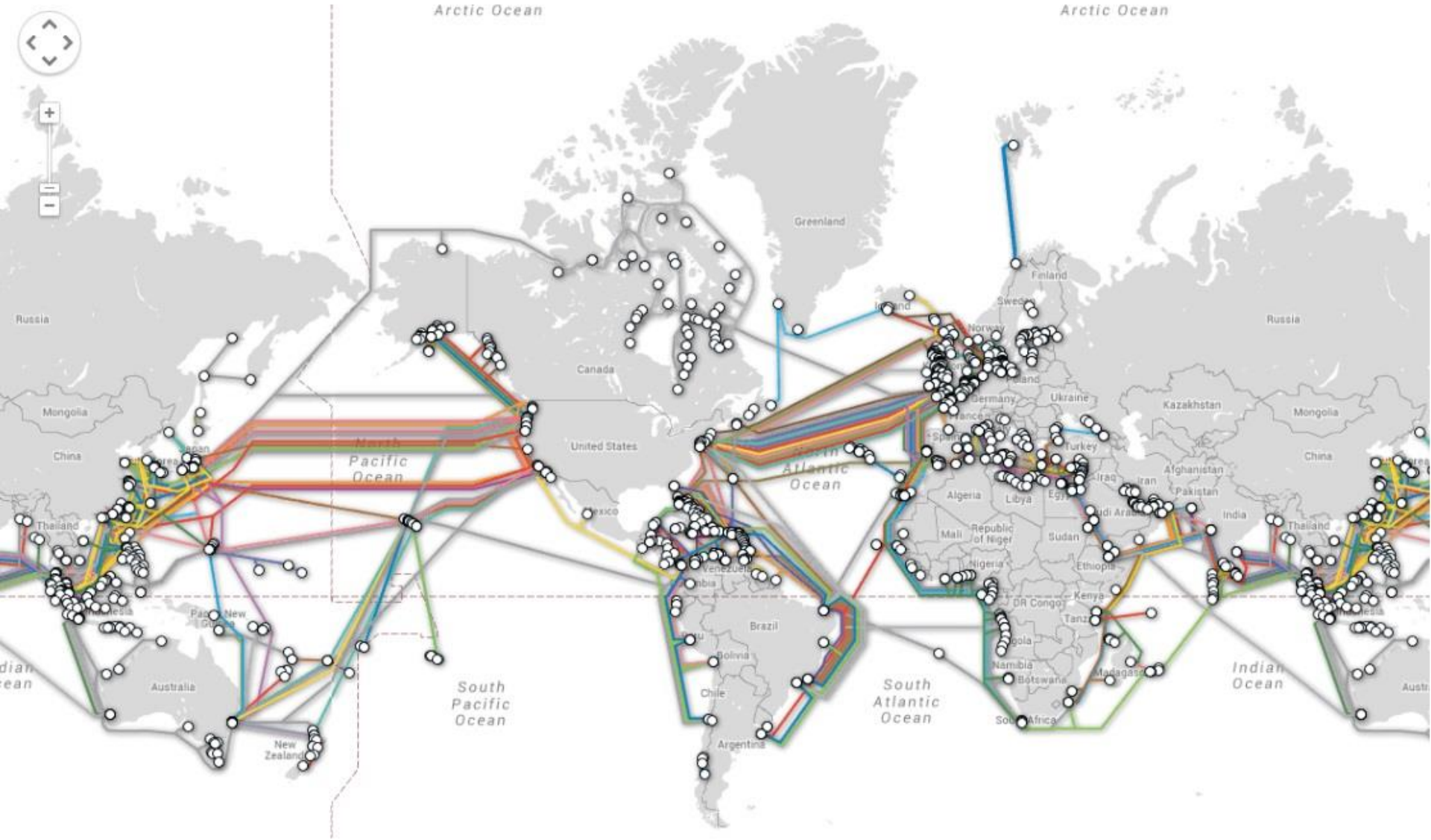
The Global Network of Computers


Mediterranean Undersea Cables
SAS-1 Active 1.3 terabits
SEA-ME-WE 4 Active 1.3 terabits
I-ME-WE Active 3.8 terabits
EIG Active 3.8 terabits
SEA-ME-WE 5 Active 24 terabits
AAE-1 Active 40 terabits
N.B. Several smaller Mediterranean cables not shown.
West Coast
SAT3/SAFE 800 gigabits Active
GLO-1 2.5 terabits Active
MainOne 10 terabits Active
ACE 12.8 terabits Active
NCSCS 12.8 terabits Active
WACS 14.5 terabits Active
SHARE 16 terabits Q1 2021
SAIL 32 terabits Active
SACS 40 terabits Active
EllaLink 72 terabits Q4 2020
Equiano 100 terabits Q4 2021
Pan African
2africa 180 terabits Q4 2023
East Coast
SEAS 320 gigabits Active
LION2 1.3 terabits Active
TEAMs 1.6 terabits Active
EASSy 10 terabits Active
Seacom 12 terabits Active
METISS 24 terabits Q3 2020
DARE1 60 terabits Q2 2020
PEACE 60 terabits Q? 2020
London England
Penmarch France
Marseille France
Monaco
Bari Italy
Palermo Italy
Sesimbra Portugal
Chipiona Spain
Asilah Morocco
Casablanca Morocco
Annaba Algeria
Bizerte Tunisia
Tripoli Libya
Marmaris Turkey
Tripoli Lebanon
Alexandria Egypt
Cairo Egypt
Suez Egypt
Altavista Canary Islands
Nouakchott, Mauritania
Dakar, Senegal
The Gambia
Cape Verde
Conakrie, Guinea
Freetown, Sierra Leone
Monrovia, Liberia
Abijan,Cote D'Ivoire
Accra,Ghana
Lome, Togo
Cotonou, Benin
Lagos, Nigeria
Douala, Cameroun
Kribi, Cameroun
Bata, Equatorial Guinea
Libreville, Gabon
to Fortaleza Brazil
Sao Tome and Principe
Pointe Noir, Congo
Muanda, DRC
Cacuaco,Angola
Sangano,Angola
Swakopmund Namibia
Yzerfontein
Melkbosstrand
Maputo Mozambique
Mtunzini
Amanzimtoti
Doha Qatar
Kalba UAE
Karachi Pakistan
Fujairah UAE
Qalhat Oman
Jeddah Saudi Arabia
Port Sudan Sudan
Massiwa Eritrea
Mocha Yemen
Djibouti
Berbera
Bossasa Somalia
Socotra
Mumbai India
Chennai India
Cochin India
Colombo SriLanka
Mogadishu Somalia
Mombasa Kenya
Dar Es Salaam Tanzania
Seychelles
Moroni Comores
Mayotte
Tamatave Madagascar
Toliara Madagascar
Libanona Madagascar
Baie du Jacobet, Mauritius
St Paul Reunion
African Undersea Cables (2023)
http://manypossibilities.net/african-undersea-cables
Version 51
Jul 2020

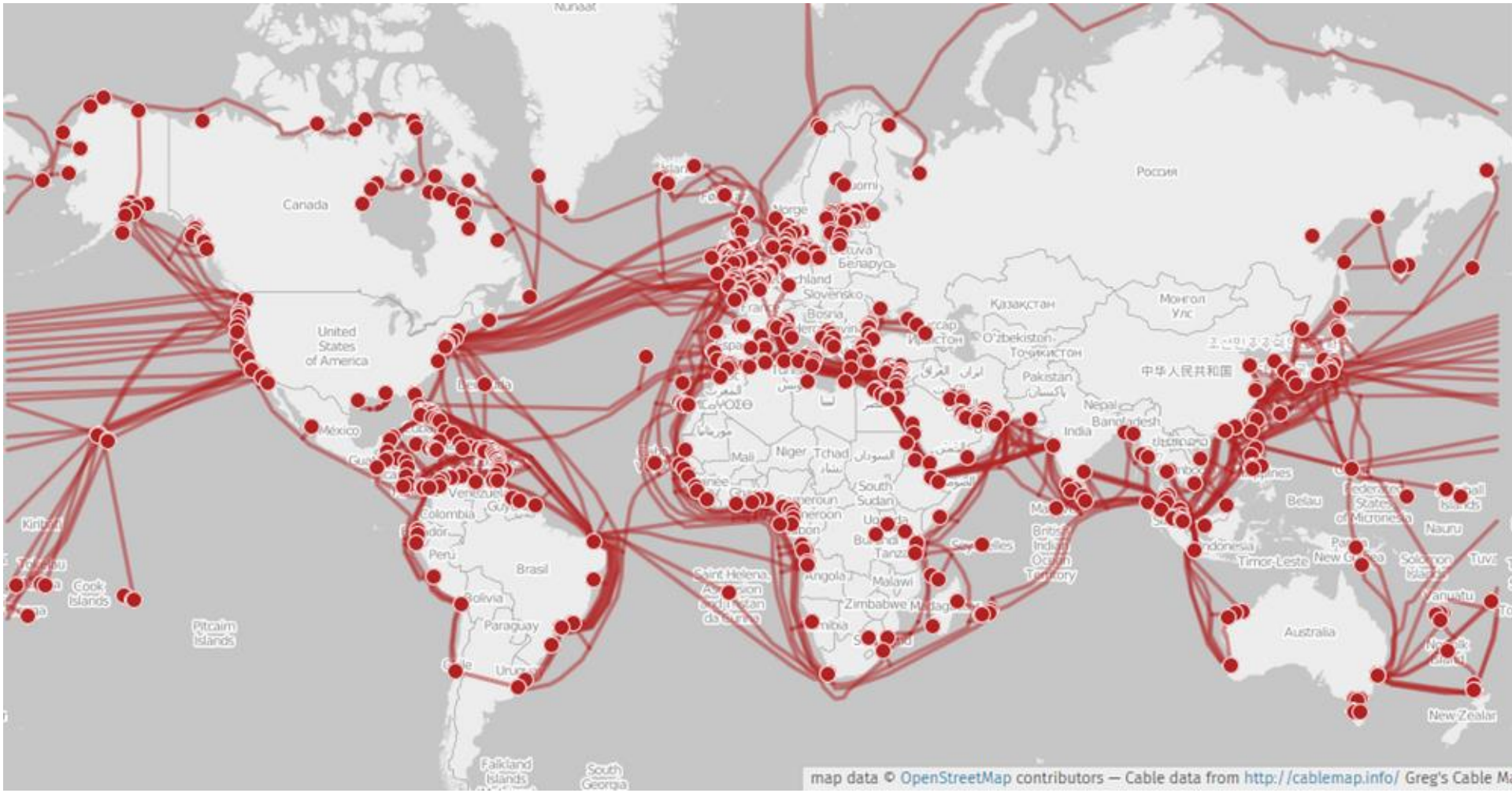
Facebook, telcos to build huge subsea cable for Africa and Middle East



Undersea cables that keep the Internet alive



Undersea cables that keep the Internet alive



Exploring The Hidden Internet



	Surface Web	Deep Web	Dark Web
Description	Content that search engine can find	Content that search engine cannot find	Content that are hidden intentionally
Known as	Visible web, Indexed web	Invisible web, Hidden web, Deep web	-
Constitutes	Web	Web	Web
Contents	Legal	Legal + Illegal	Illegal
Information found	4%	96%	-
Browsers	Google chrome, Mozilla firefox, opera	-	TOR browser

Connection to Internet

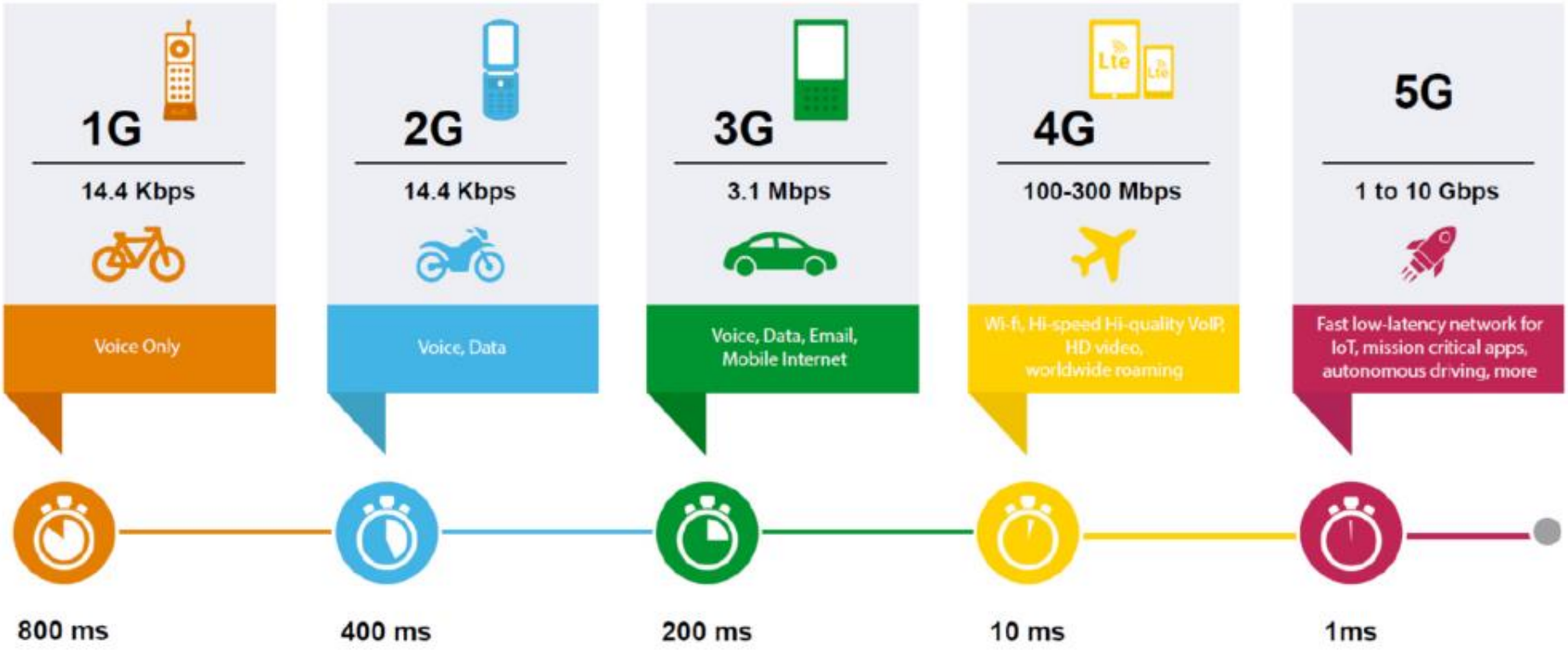
- Mobile broadband - USB modem (dongle)/MiFi
- Home Broadband - Digital Subscriber Line (DSL)
- WiFi Hotspot
- Cellular networks

Mobile Broadband

- Wireless Internet access through a portable modem.



EVOLUTION OF 1G TO 5G



4G Devices

MIFI



USB Modem



Router



Home broadband

- Digital Subscriber Line (DSL)
  - Faster than mobile broadband
  - Requires a fixed telephone line

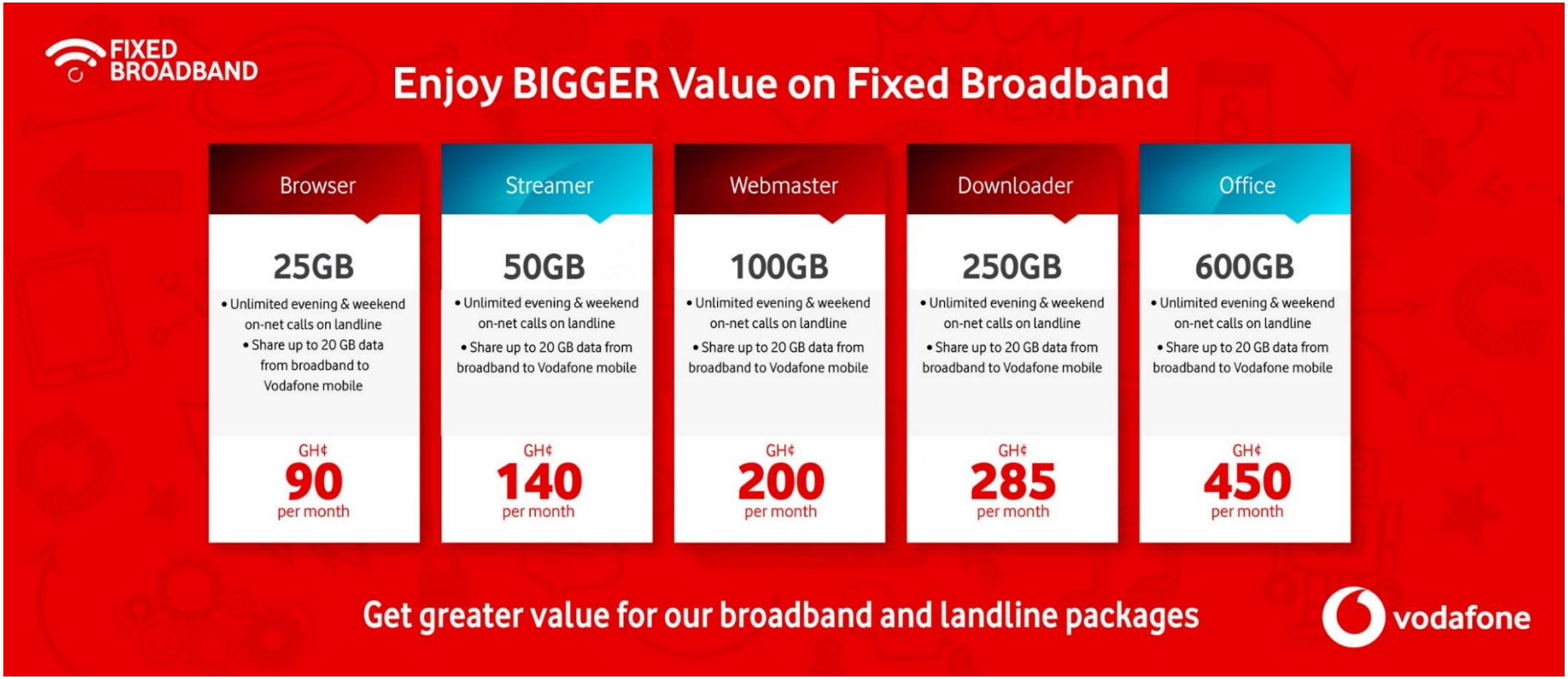






Home broadband

- Vodafone Fixed broadband



Connecting to the Internet

- An **Internet service provider** (**ISP**) is a business that provides individuals and organizations access to the Internet free or for a fee
- Bandwidth represents the amount of data that travels over a network
  - Megabyte (MB)
  - Gigabyte (GB)

Internet Service Providers

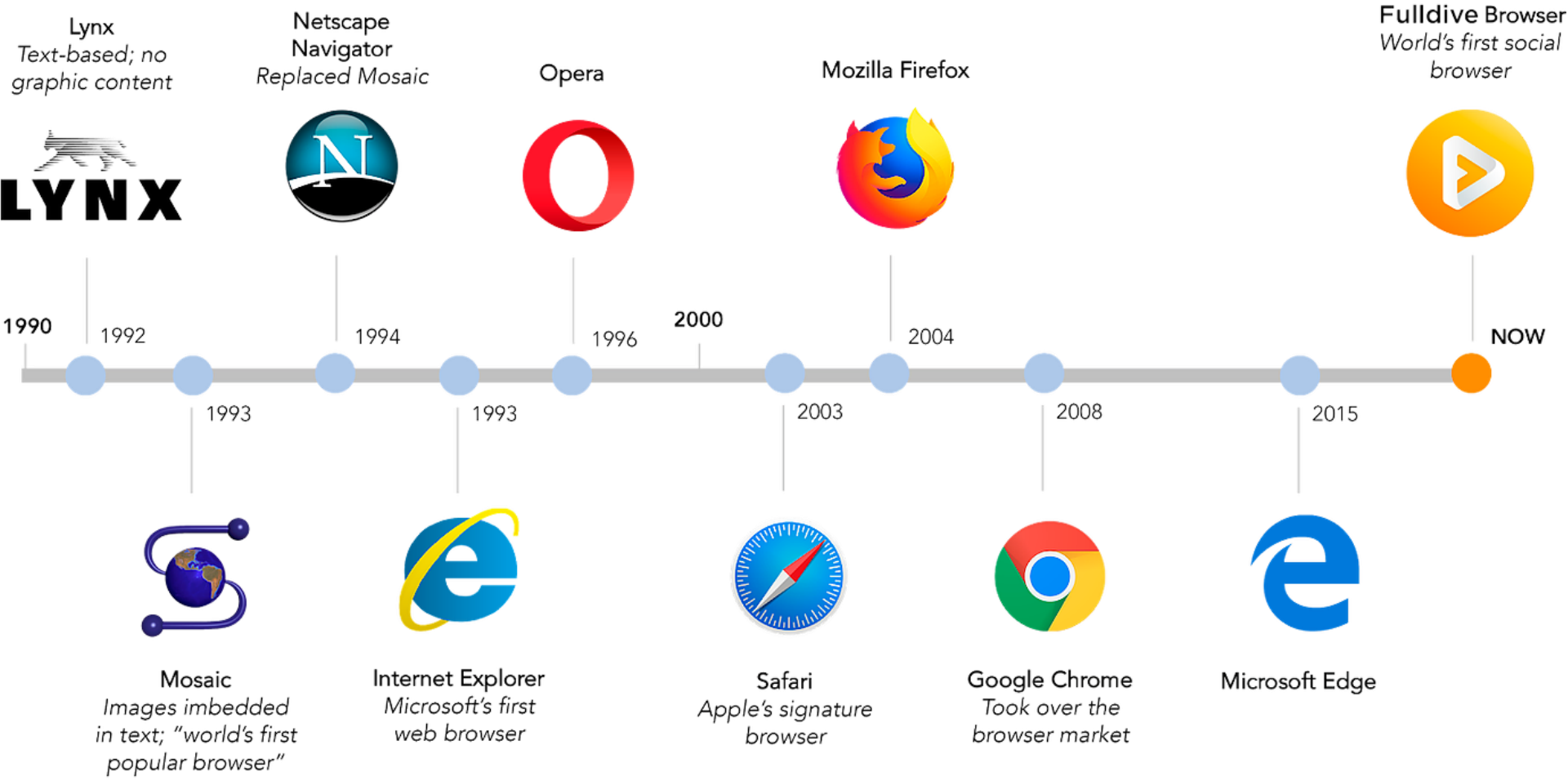
- List of companies:
  - All telcos
  - Busy internet
  - Zipnet
  - iBurst
  - Main one
  - Internet Ghana
  - Glo-1
  - Surline

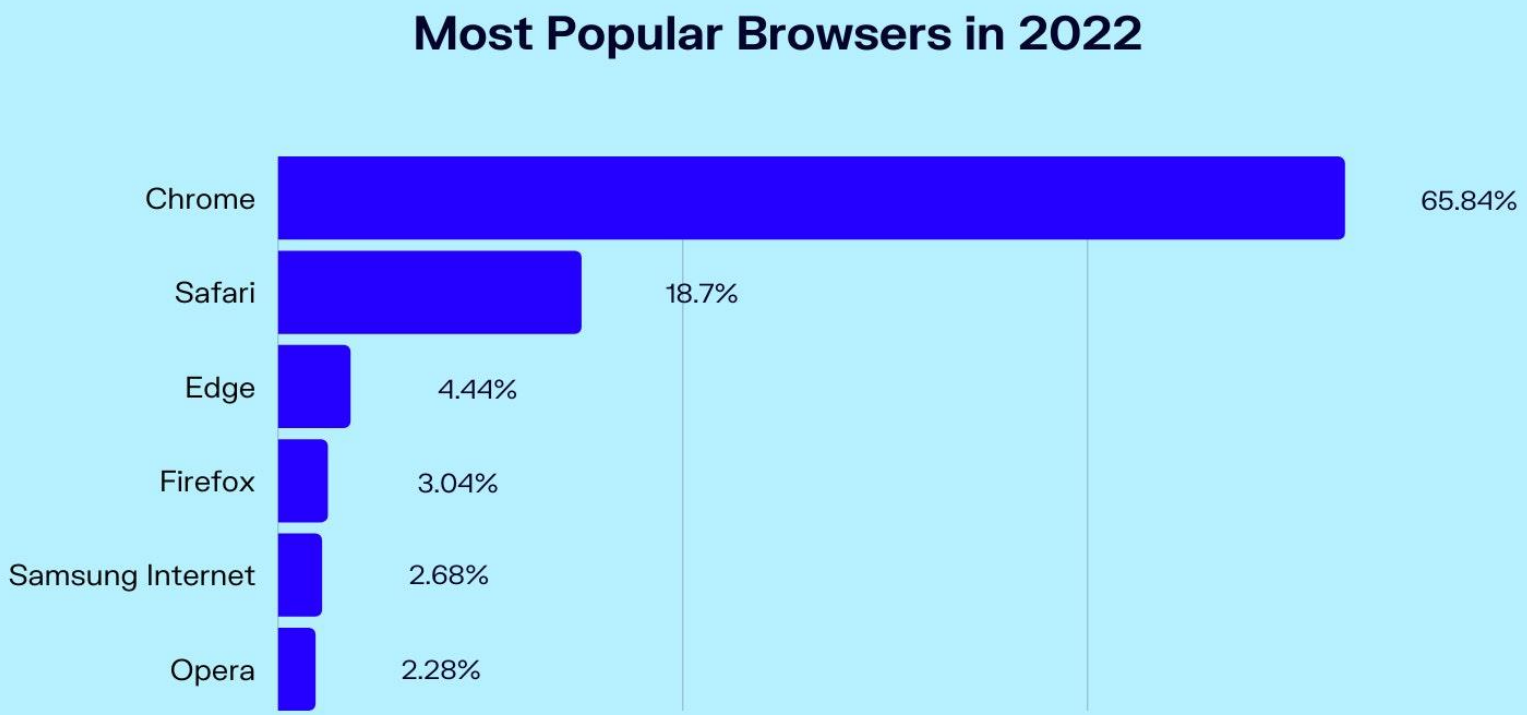
Connecting to the Web

- Web browser is software that allows you to locate, view, and navigate the Web.



Browser Timelines





Source: Gs.statcounter.com

OBERLO

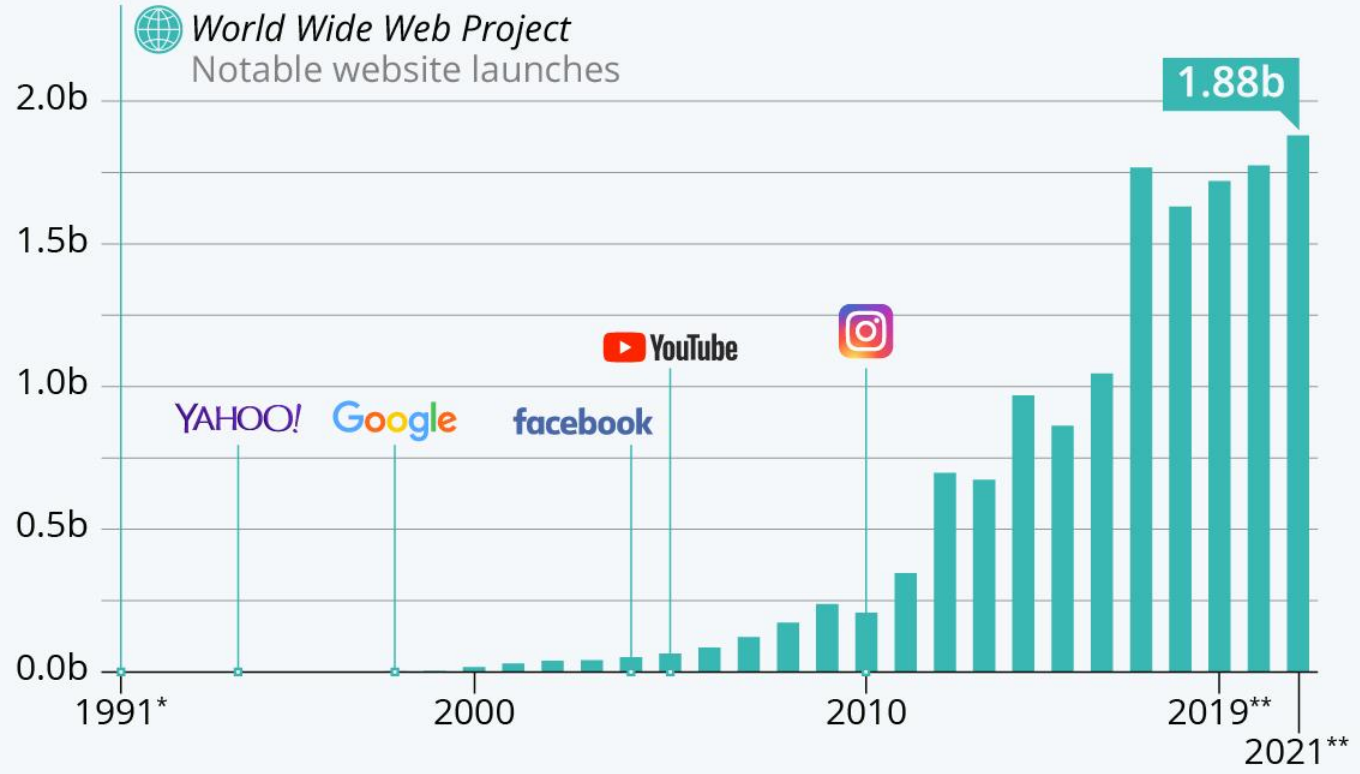
Getting Around the Web

- Web sites
- URLs
- Hyperlinks
- Favorites and Bookmarks



How Many Websites Are There?

Number of websites online from 1991 to 2021



* As of August 1, 1991.

** Latest available data for 2019: October 28, for 2020: June 2, for 2021: August 6.

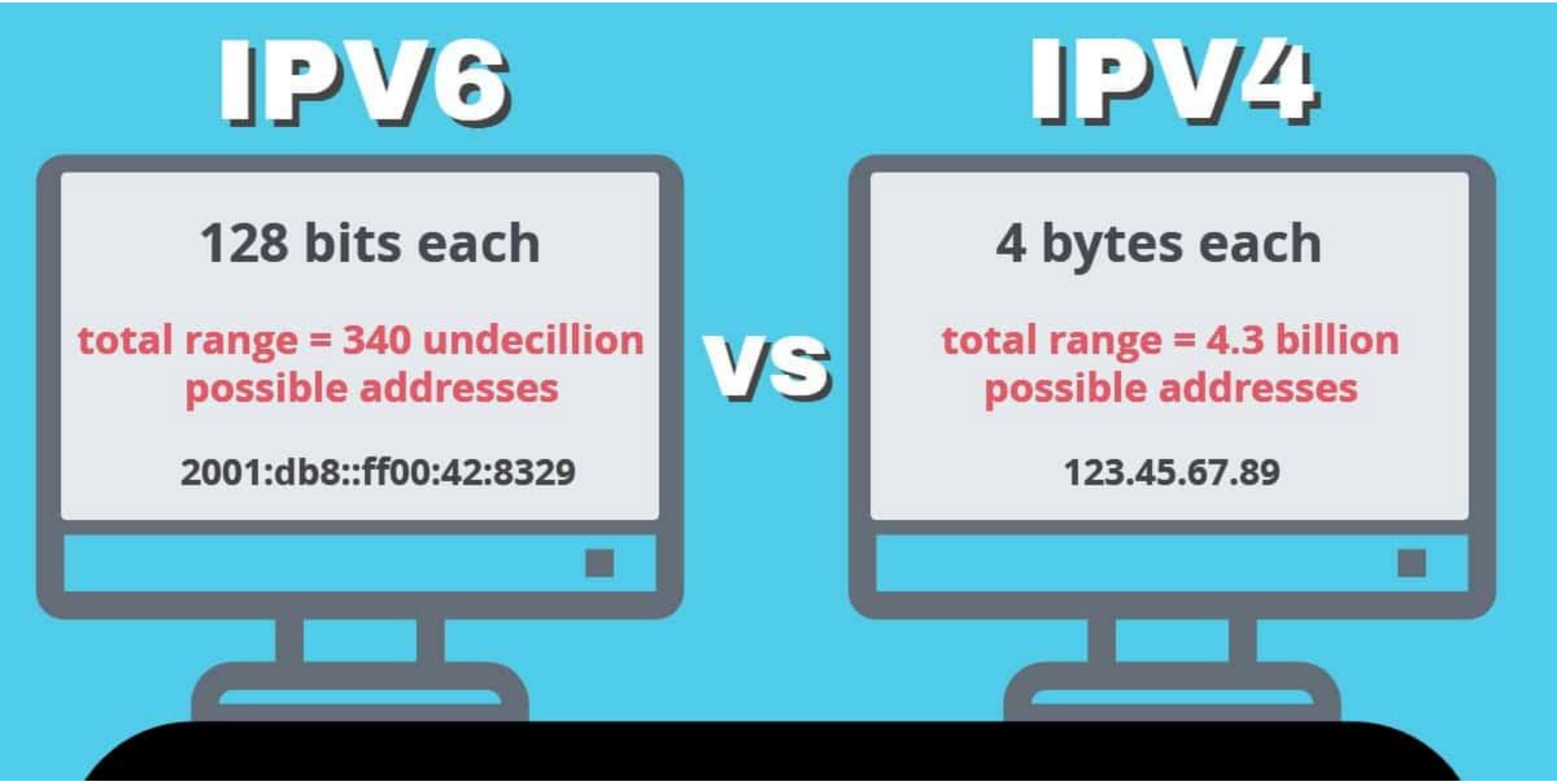
Source: Internet Live Stats

What is an IP address?

- To check your IP Address on Android Phone
  1. Go to setting
  2. About device
  3. Status
  4. Locate IP address

Internet Addressing and Architecture

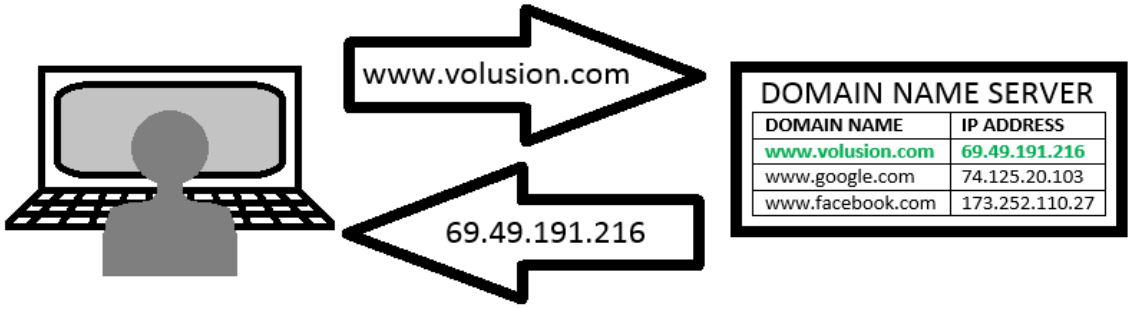
- Internet Protocol (IP) Address
- Every computer on the Internet is assigned a unique IP.
- To access a computer on the internet, you only have to specify its IP address.
- An example of IP address: **207.46.250.119**, **10.0.0.1**
- The numbers ranges from 0 – 255


IPV6
128 bits each
total range = 340 undecillion possible addresses
2001:db8::ff00:42:8329
VS
IPV4
4 bytes each
total range = 4.3 billion possible addresses
123.45.67.89

	Internet Protocol version 4 (IPv4)	Internet Protocol version 6 (IPv6)
Deployed	1981	1999
Address Size	32-bit number	128-bit number
Address Format	Dotted Decimal Notation: 192.149.252.76	Hexadecimal Notation: 3FFE:F200:0234:AB00: 0123:4567:8901:ABCD
Prefix Notation	192.149.0.0/24	3FFE:F200:0234::/48
Number of Addresses	2^{32} = ~4,294,967,296	2^{128} = ~340,282,366, 920,938,463,463,374, 607,431,768,211,456

Internet Addressing and Architecture

- Domain Name and Domain Name Systems
- Remembering the IP address is difficult hence an english-like description of IP hence **domain names.**
- Domain Name is the English-like name that corresponds to an IP address.
- **Domain Name system (DNS)** converts domain names to IP addresses.



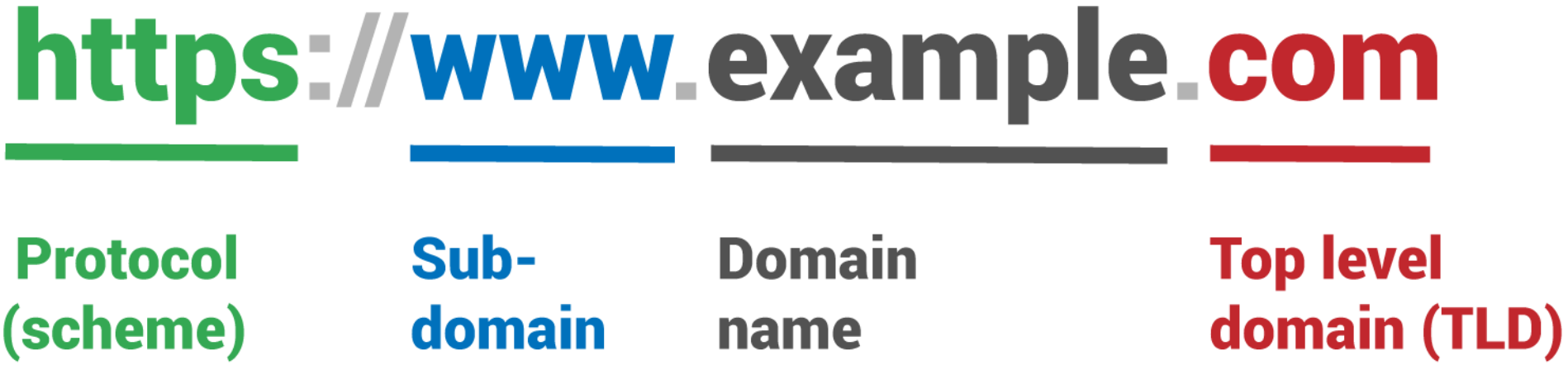
Internet architecture and governance

- No formal management: **IAB, ICANN, W3C**
- **IAB** - Internet Architecture Board
  - Internet Society that overseer the technical evolution of the Internet.
- **ICANN** - Internet Corporation for Assigned Names and Numbers
  - a private (non-government) non-profit corporation with responsibility for IP address space allocation, protocol parameter assignment, domain name system management etc.
- **W3C -** World Wide Web Consortium
  - develop standards for the World Wide Web.

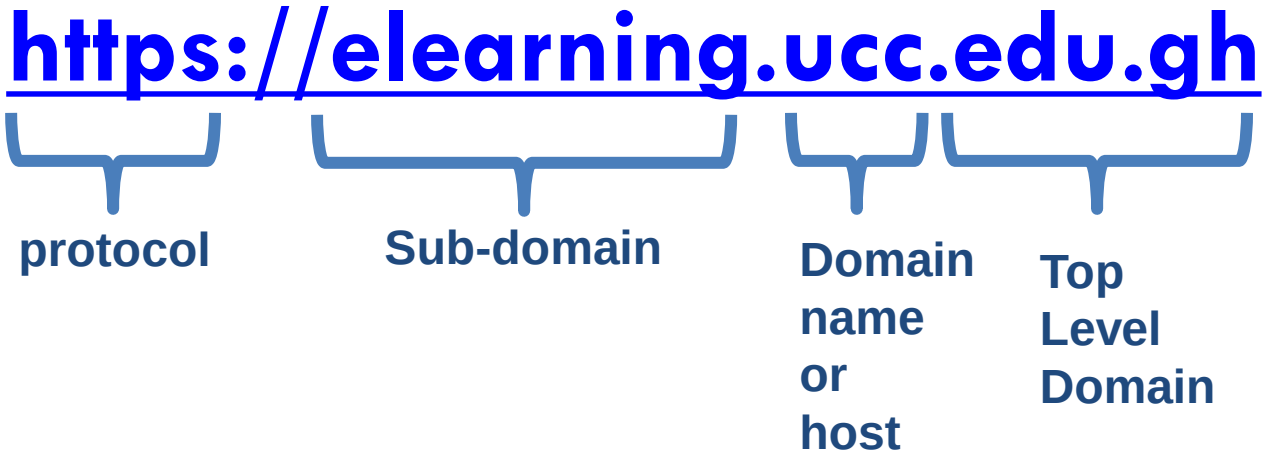
Uniform Resource Locator (URL)

- URL is the address of a web page or the address of a resource on the Internet.
- Example URLs:
  - www.ucc.edu.gh
  - https://dcsit.ucc.edu.gh/
  - www.ghana.gov.gh
  - www.ges.gov.gh
  - www.vodafone.com.gh
  - www.google.com
  - www.google.com.gh

Parts of web address or URL



Identify the paths of this URL:



Identify the paths of this URL

https://www.ucc.edu.gh/computing

Identify the parts of the email address

dodzi@stu.ucc.edu.gh

eattipoe@ucc.edu.gh

Current Top-Level Domains

TLD	Meaning
.com	Can be used by anyone (commercial)
.edu	Academic institution
.gov	Governments
.info	Information Service Providers
.net	Networking Organisation
.org	Organisations (often non-profit)
.gh .com.gh or .edu.gh	Country domain

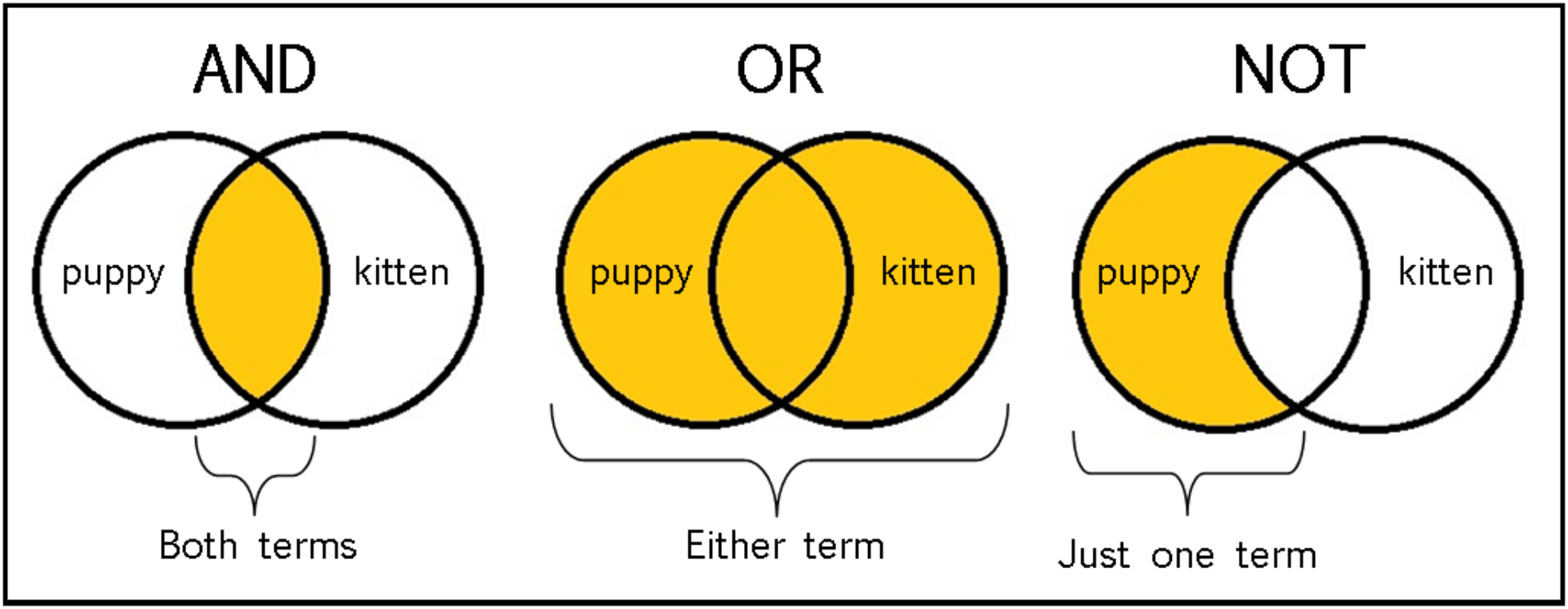
Types of Websites

Portal	Search engines	News	Personal
E-commerce	Blogs	Wiki	Social networks
Educational	Entertainment	Informational	Web applications

Search Engine

- A web search engine is software that finds websites, webpages, images, videos, news, maps, and other information related to a specific topic.

Boolean Search Operators



What are the positive and negative impact of social networking on students?

Content Management System (CMS)

- CMS is software that helps users create, manage, and modify content on a website without the need for specialized technical knowledge.
- In simpler language, a CMS is a tool that helps you build a website without needing to write all the code from scratch (or even know how to code at all).

Popular content management systems

- WordPress
- Joomla
- Drupal
- Magento (for eCommerce stores)
- Squarespace
- Wix
- ExpressionEngine
- Alfresco

Criteria for Evaluating the Accuracy/Validity of a Website

- Authority and Affiliation
- Audience
- Accuracy
- Objectivity
- Currency

Criteria for Evaluating the Accuracy/Validity of a Website

- **Authority and Affiliation**
  - Authority reveals that the person, institution or agency responsible for a site has the qualifications and knowledge to do so.
  - Contact information should be clearly provided: e-mail address, snail mail address, phone number, and fax number.
  - Credentials: the author should state qualifications, credentials, or personal background that gives them authority to present information.
  - Check to see if the site supported by an organization or a commercial body.

Criteria for Evaluating the Accuracy/Validity of a Website

- **Audience**
  - Who is the website's intended audience? Academics? The general public? School children?
  - Does it appropriately address the target audience?
  - Is it relevant for your assignment or research?

Criteria for Evaluating the Accuracy/Validity of a Website

- **Accuracy**
  - Is it free from spelling errors?
  - Is the text well-written and grammatically correct?
  - Has the content been through an editing process or been peer reviewed?
  - Has the author included a bibliography?
  - Are the sources cited reliable and can they be verified elsewhere?
  - Are research methodologies adequately explained?

Criteria for Evaluating the Accuracy/Validity of a Website

- **Objectivity**
  - Does the author present objective arguments or make it clear when he is expressing biased opinions?
  - Are other points of view explored?
  - Is it a personal website? Does it express personal opinions?
  - Is the website part of a commercial organization, a political party or an organization with a specific agenda? If yes, question the motives for publishing the information.
  - Does the website promote a biased viewpoint?

Criteria for Evaluating the Accuracy/Validity of a Website

- **Currency**
  - Can you tell when the information was published?
  - Is the information up to date?
  - How frequently is the website updated?
  - Are the links up to date and working?

E-commerce

- Three major types of e-commerce
  - **Business-to-consumer (B2C)**
    - Example: amazon.com, Aliexpress.com, KiKuu.com, jumia.com.gh
  - **Business-to-business (B2B)**
    - Example: Alibaba.com
  - **Consumer-to-consumer (C2C)**
    - Example: eBay, tonaton

E-commerce apps







Mobile Payment Apps

- Bank Apps
- Mobile Money apps
- Third party apps
  - Expresspay
  - Slydepay
  - Zeepay
  - Hubtel
  - Mazzuma
  - Paypal*

E-commerce revenue models

- Advertising
- Sales
- Subscription
- Free/Freemium
- Transaction fee
- Affiliate

How to generate revenue from the web

- How to generate revenue from youtube.
- Enable AdSense
- Your youtube videos with 1000 views you earn **$7.6**
- **Gangnam Style** had 1 billion views and was paid **$7.8 million**

Music Streaming Services

Streaming Platform	Amount per stream ($)	Amount per 1000 streams ($)
Tidal	0.011	11.00
Apple Music	0.0073	7.30
Spotify	0.0018	1.80
Deezer	0.0011	1.10
Google Play Music	0.0067	6.70
Amazon Music	0.0082	8.20
YouTube Music	0.0029	2.91

Video Streaming Services

showmax

amazon Prime

NETFLIX

IROKO TV

Google Play Movies & TV

Netiquette

- **Netiquette**, which is short for Internet etiquette, is the code of acceptable behaviors users should follow while on the Internet; that is, it is the conduct expected of individuals while online.

Netiquette Guidelines for Online Communications

Golden Rule: Treat others as you would like them to treat you.

Be polite. Avoid offensive language.

Avoid sending or posting *flames*, which are abusive or insulting messages. Do not participate in *flame wars*, which are exchanges of flames.

Be careful when using sarcasm and humor, as it might be misinterpreted.

Do not use all capital letters, which is the equivalent of SHOUTING!

Use **emoticons** to express emotion. Popular emoticons include:

:) Smile :| Indifference :o Surprised :(Frown :\ Undecided ;) Wink

Use abbreviations and acronyms for phrases:

BTW	by the way	IMHO	in my humble opinion	FWIW	for what it's worth
FYI	for your information	TTFN	ta ta for now	TYVM	thank you very much

Clearly identify a spoiler, which is a message that reveals an outcome to a game or ending to a movie or program.

Be forgiving of other's mistakes.

Read the *FAQ* (frequently asked questions), if one exists.

Research on the following

- Good online behaviour
- Rules for email etiquette